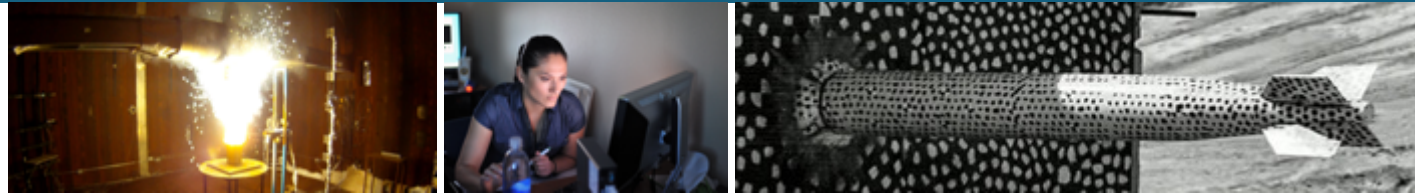


Embedded Model Error Quantification and Propagation in Physical Systems



PRESENTED BY

Khachik Sargsyan, 8351

CIS External Review, August 26-29, 2019
SAND 2019-9724 C

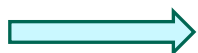


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Algorithmic talk demonstrating broad application impact

- Overview

- **Model error**



- Definition
- Current state
- What we do

- Applications



- Climate land model (BER+ASCR)
- Chemistry (BES)
- Fusion science (FES+ASCR)
- Thermodynamics (EERE)
-
- Turbulence modeling



- Summary/future

This work has been funded by DOE/DOD, including fundamental research and applications



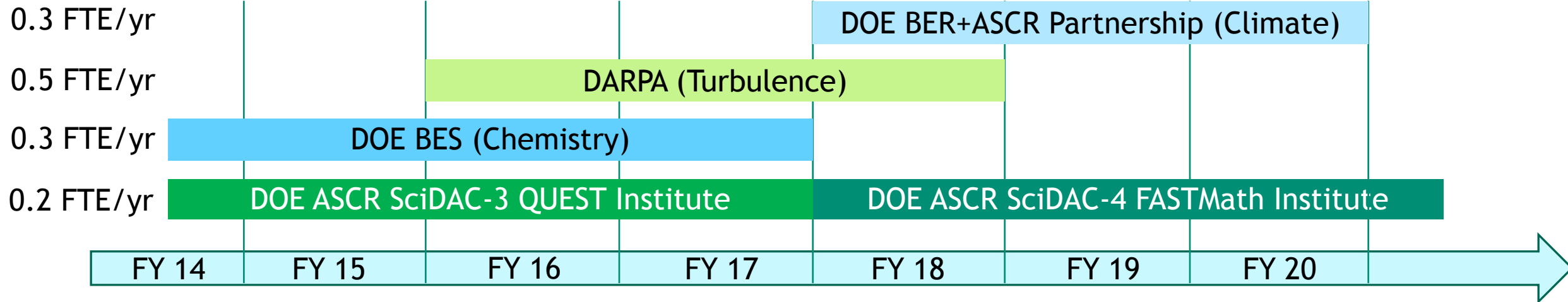
model-error work
incl. collaborators



U.S. DEPARTMENT OF
ENERGY
Office of Science



SciDAC
Scientific Discovery
through
Advanced Computing



Habib Najm, Cosmin Safta, Tiernan Casey, James Oreluk, Bert Debusschere (SNL)



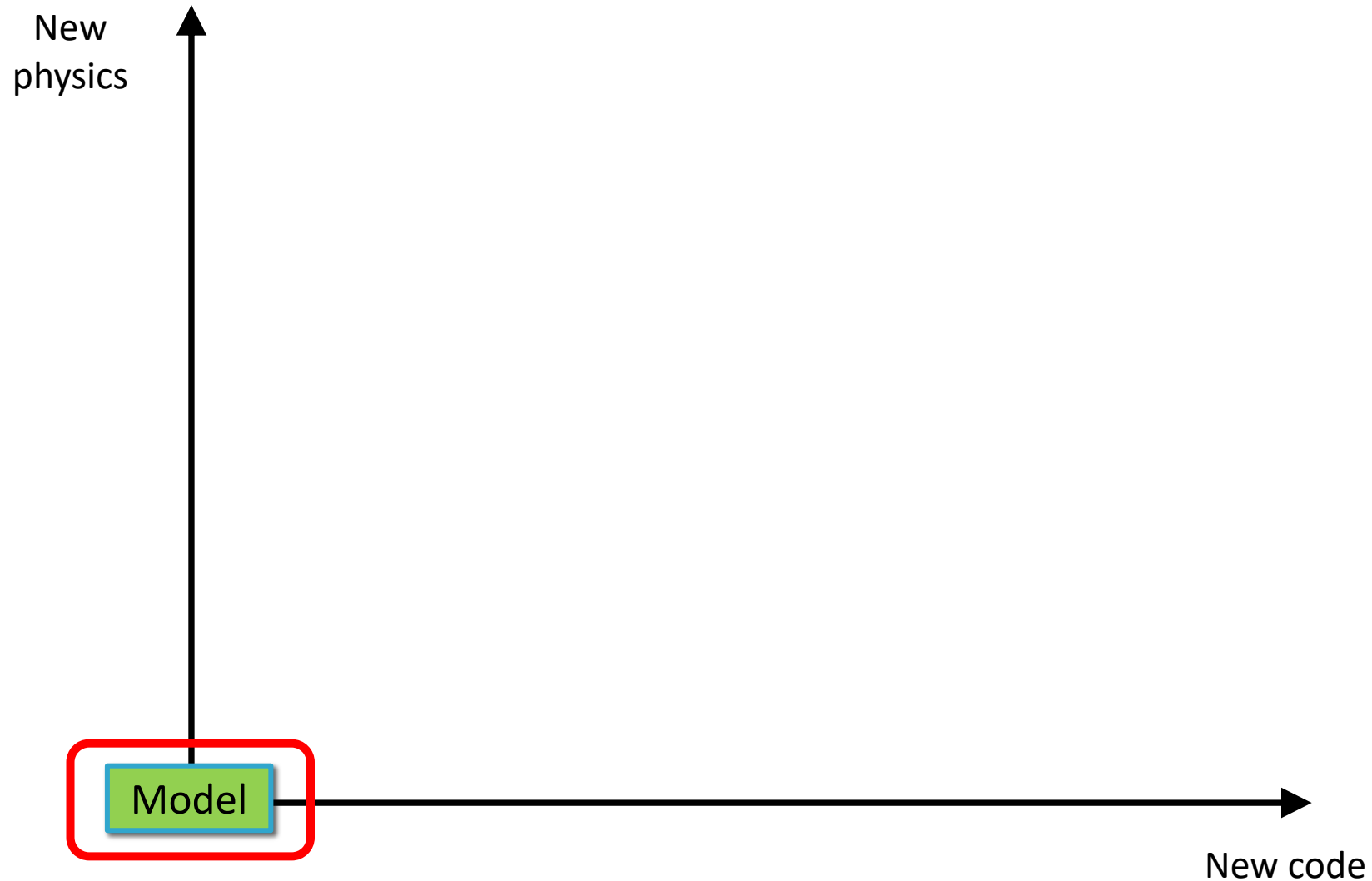
Daniel Ricciuto (ORNL), Jason Bender (LLNL)



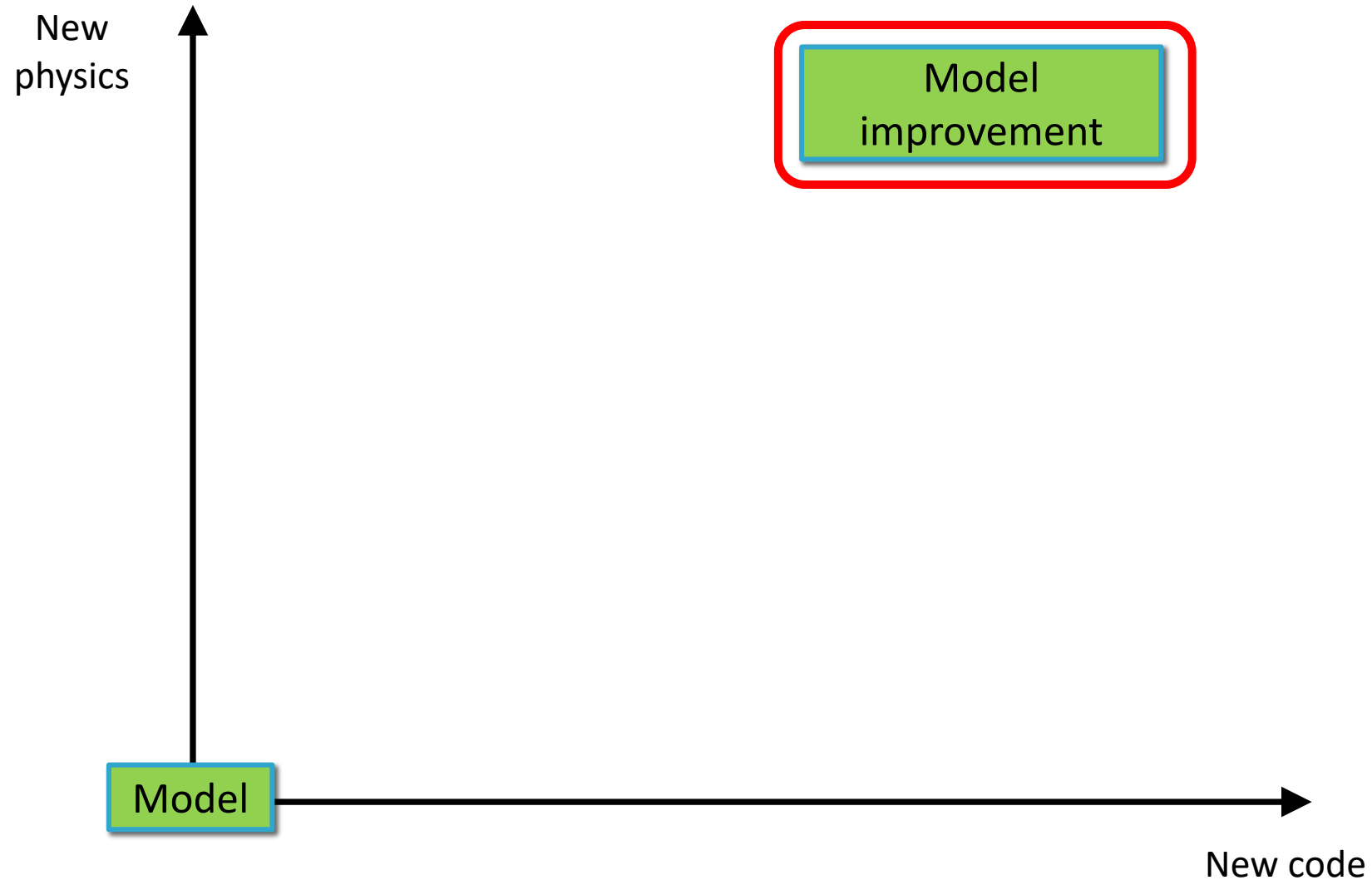
Youssef Marzouk, Chi Feng (MIT), Roger Ghanem (USC), Xun Huan (UMichigan)



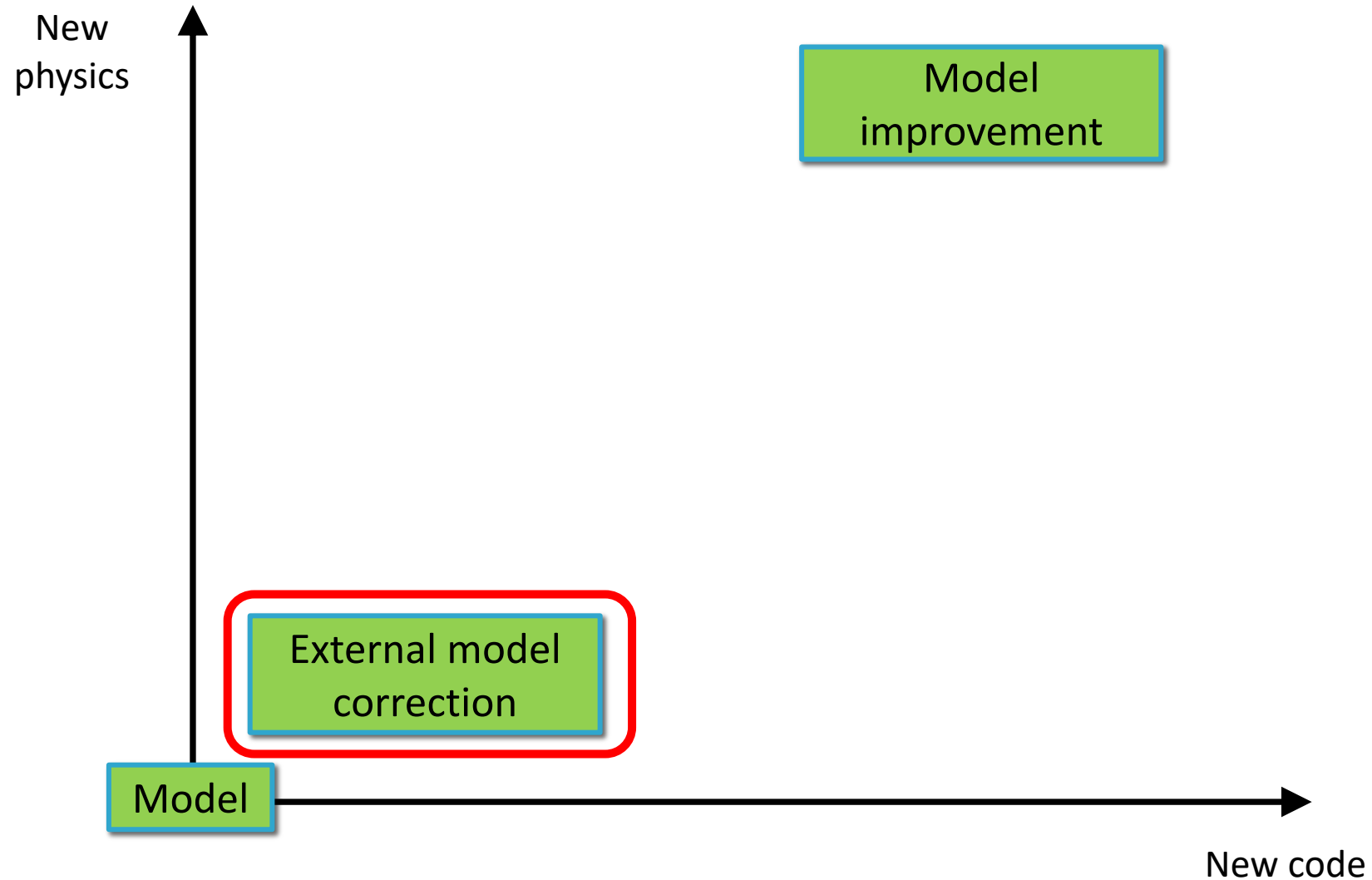
**To improve computational models of physical phenomena,
one targets improvements in the physics and the code**



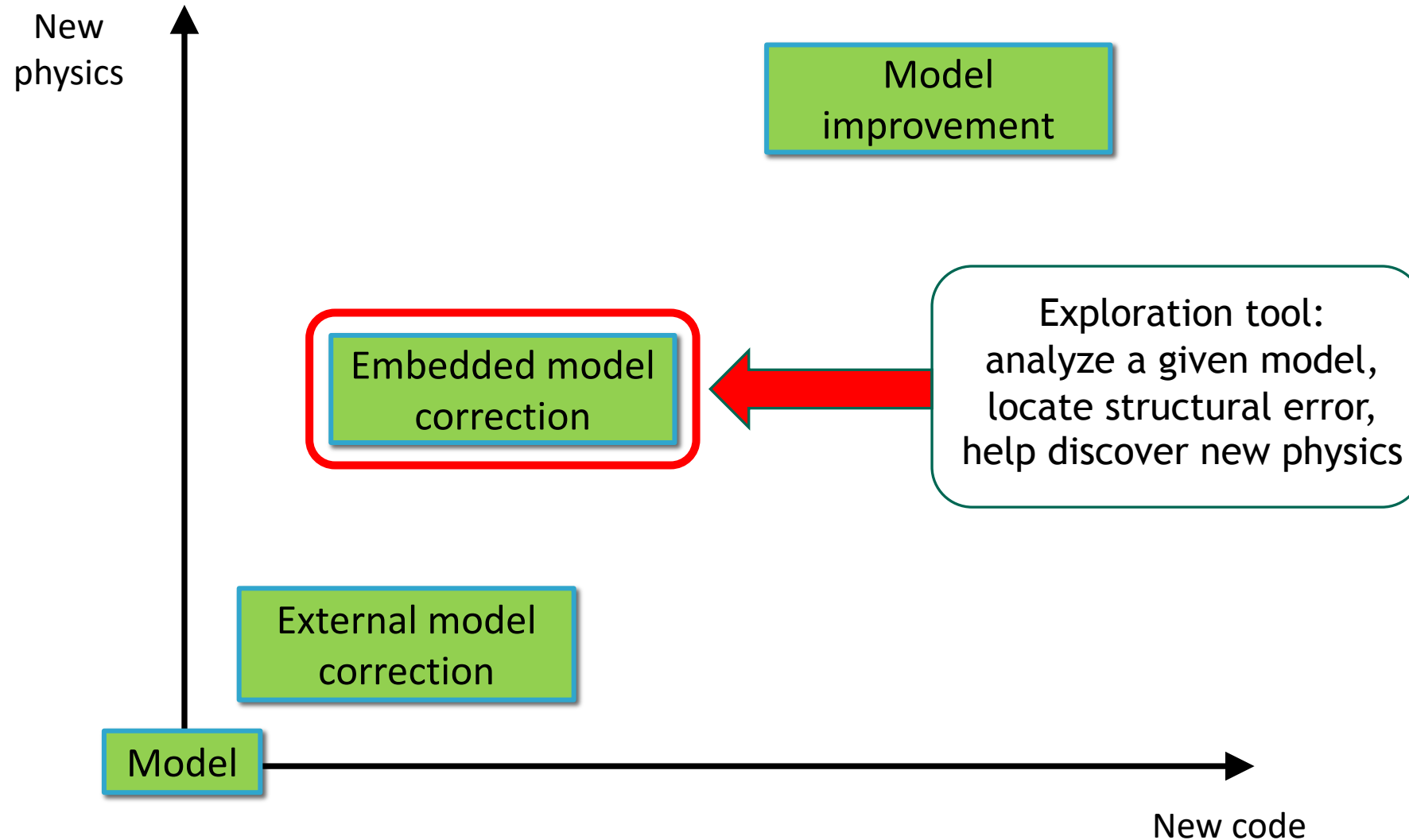
Ideal (but costly) situation: domain-specific 'intrusion' into the code



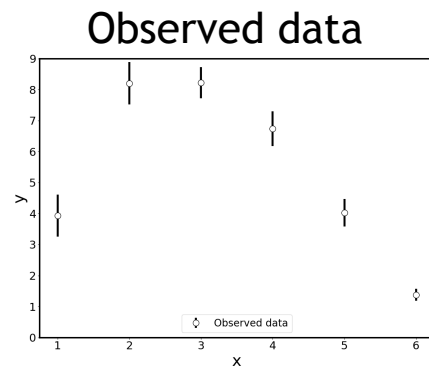
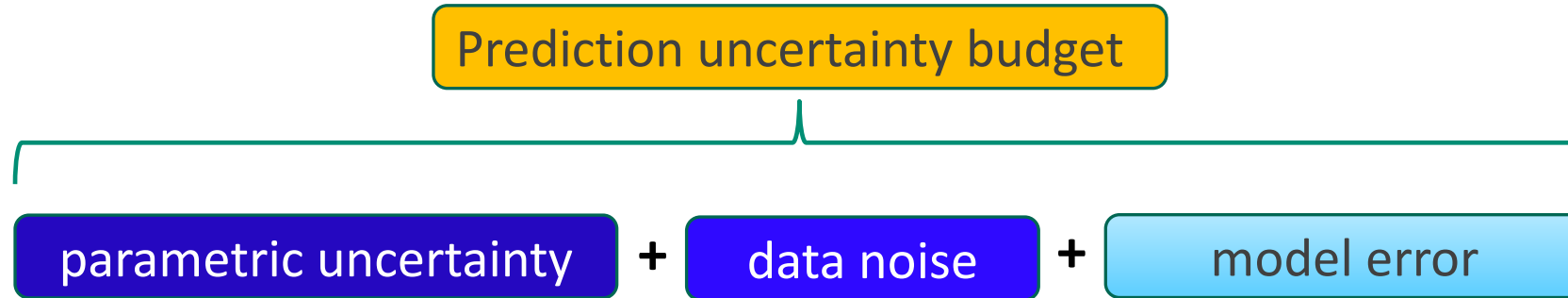
There is mature work in *statistical* model corrections, without physical interpretation



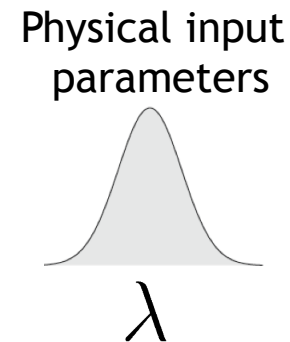
**We develop approach for embedded statistical correction,
that is driven by the physics but does not require code change
(augments the existing code with stochasticity)**



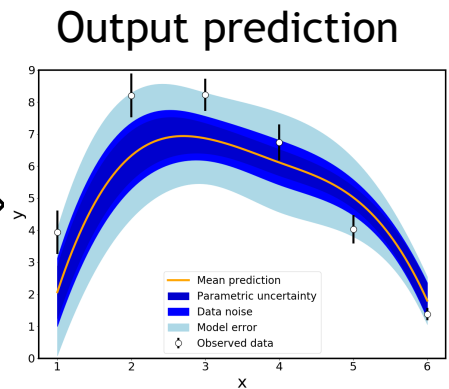
In order to improve or correct the model, we need to quantify model error



Meas. model
 $y(x)$



Comp. model
 $f(x; \lambda)$



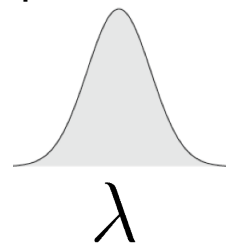
In UQ, we know how to incorporate parametric uncertainty

Prediction uncertainty budget

parametric uncertainty

Cosmin Safta's talk from 2018 CIS ERB:
algorithmic advances to address
high-dimensionality of λ and
non-linearity of $f(x; \lambda)$

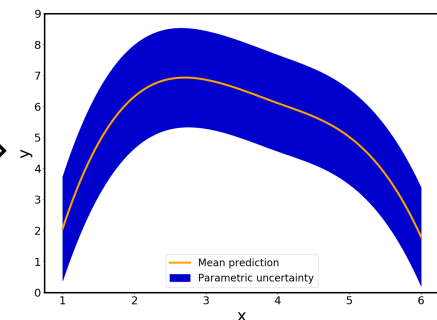
Physical input
parameters



Comp. model

$$f(x; \lambda)$$

Output prediction



Forward modeling

In UQ, we know how to incorporate parametric uncertainty and data noise

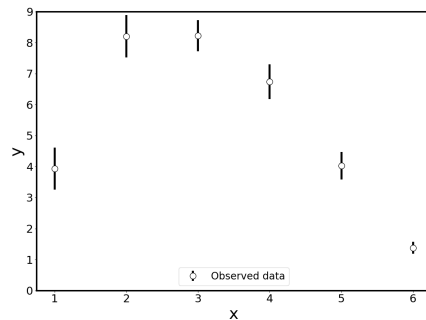
Prediction uncertainty budget

parametric uncertainty

+

data noise

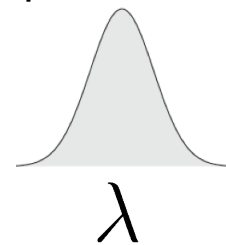
Observed data



Meas. model

$y(x)$

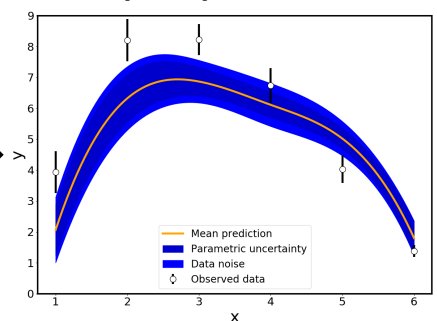
Physical input parameters



Comp. model

$f(x; \lambda)$

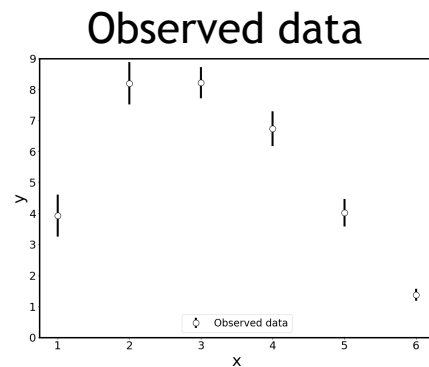
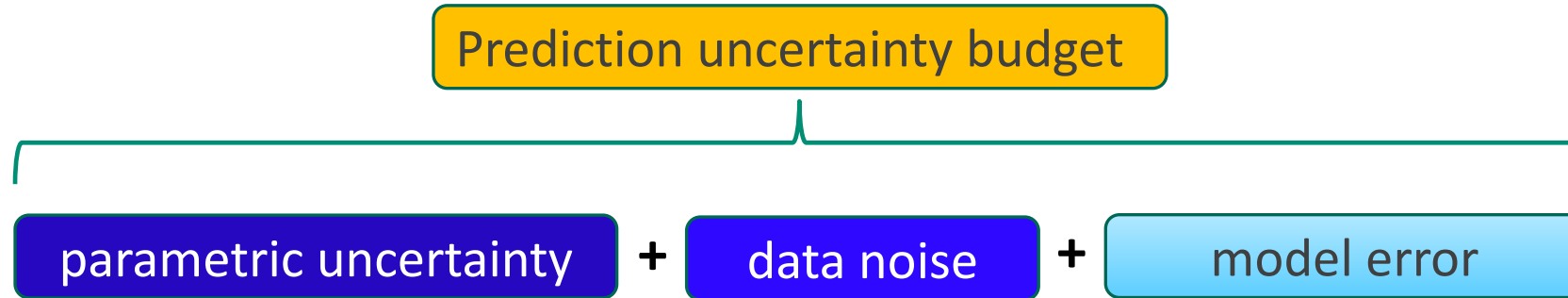
Output prediction



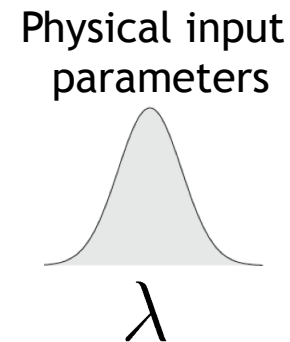
Inverse modeling

Forward modeling

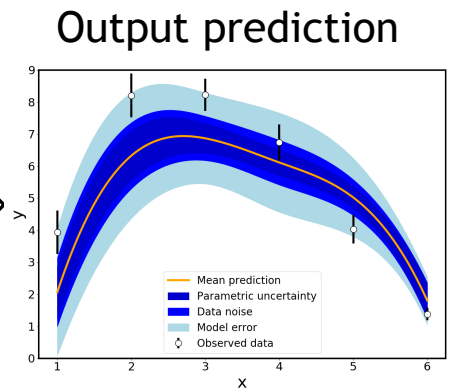
In order to improve or correct the model, we need to quantify model error



Meas. model
 $y(x)$



Comp. model
 $f(x; \lambda)$



Model error is often the dominant component of predictive uncertainty



Ignoring model error leads to

- Biased parameter estimation
- Overconfident predictions

$$\begin{array}{ccccc} \text{Data} & & \text{Model} & & \text{Data noise} \\ y(x_i) & = & f(x_i; \lambda) & + & \epsilon_i \end{array}$$

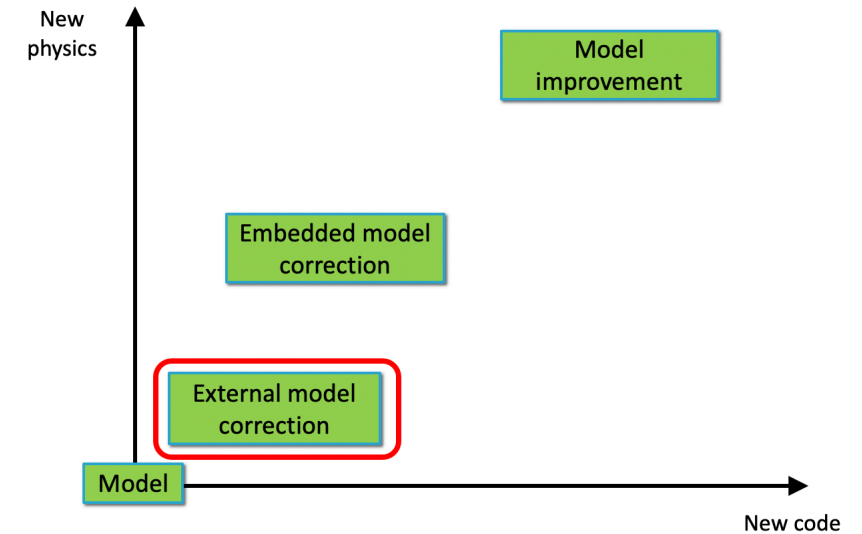
Representing and estimating model error is useful for

- Reliable computational predictions
- Model comparison, selection
- Scientific discovery and model improvement:
 - *“is it worth resolving details, or just parameterize empirically?”*
- Optimal resource allocation:
 - *“do I improve my model (e.g. high-res), or run more simulations?”*

Conventional approach from statistical literature: **external** correction to the model



$$\begin{array}{ccccccc} \text{Data} & & \text{Model} & & \text{Model error} & & \text{Data noise} \\ y(x_i) = & & f(x_i; \lambda) & + & \delta(x_i) & + & \epsilon_i \\ & & \underbrace{\hspace{10em}} & & & & \\ & & \text{Truth} & & g(x_i) & & \end{array}$$



- Explicit Gaussian Process representation of model error [Kennedy-O'Hagan, 2001]
- See also [Higdon et. al, 2004], [Bayarri et. al, 2007]
- Usage: too many to cite
- Variants exist: multiplicative noise, non-linear maps etc.
- Implemented in Dakota – highly mature procedure

Conventional approach from statistical literature: **external** correction to the model



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**DAKOTA**

Explore and predict with confidence.

See Kathryn Maupin's poster

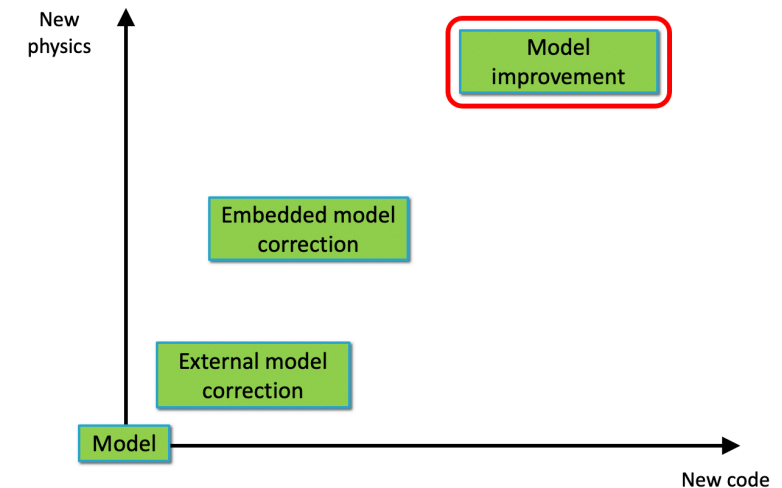
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Embedded model error corrections have gained recent interest



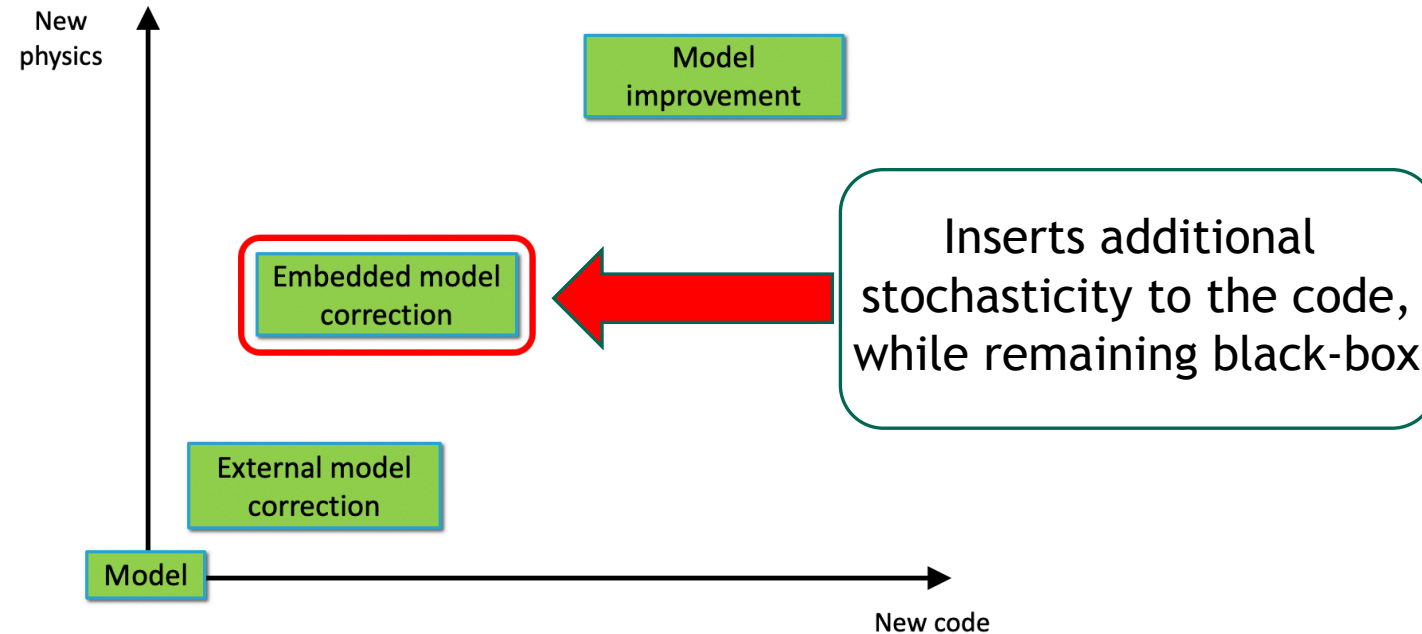
Domain scientists improve the models by problem-specific internal adjustments

- Validating extrapolative predictions [Oliver and Moser, 2011], [Oliver et. al, 2014]
- Engineering/statistical adjustment [Joseph and Melkote, 2009]
- Stochastic operator [Morrison et. al, 2016], [Sondak et. al, 2017]
- Additive corrections to submodels [Strong et. al, 2011]
- Field inversion and machine learning [Duraismy et. al, 2015-]
- Turbulence anisotropy perturbation [Emory et. al, 2011]
- Random field correction [Brown and Atamturktur, 2016]
- Hierarchical mixture model [Feng, 2017]
- Parameter inflation [Pernot et. al, 2017]
- Hierarchical stochastic model [Wu et. al, 2017]
- Dynamic discrepancy [Bhat et. al., 2017]
- Hybrid correction [He and Xiu, 2016]



$$y(x_i) = \tilde{f}(x_i; \lambda, \delta(x_i)) + \epsilon_i$$

We have developed a non-intrusive way to embed model error within the model



Embed model error where model error (i.e. key modeling assumptions) happens:

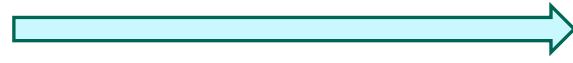
- Specific submodel phenomenology
- Modified transport law, material property, turbulent constant

- Naturally preserves physics
- Disambiguates from data noise
- Allows prediction of non-observable QoIs
- Does not vanish with more data

We have developed a non-intrusive way to embed model error within the model



External model correction



$$y(x_i) = f(x_i; \lambda) + \delta(x_i) + \epsilon_i$$

Embedded model correction

Intrusive

$$y(x_i) = \tilde{f}(x_i; \lambda, \delta(x_i)) + \epsilon_i$$

change the code

Non-intrusive
(black-box)

$$y(x_i) = f(x_i; \lambda + \delta(x_i)) + \epsilon_i$$

change inputs to the code

Embed model error where model error (i.e. key modeling assumptions) happens:

- Specific submodel phenomenology
- Modified transport law, material property, turbulent constant

- Naturally preserves physics
- Disambiguates from data noise
- Allows prediction of non-observable QoIs
- Does not vanish with more data

Polynomial chaos (PC) expansion is employed as a model error representation

$$f\left(x; \lambda + \sum_k \overset{\text{PC}}{\alpha_k} \Psi_k(\xi)\right) \overset{\text{NISP}}{\simeq} \sum_k \overset{\text{physical}}{f_k(x; \lambda)} \overset{\text{model-error}}{\alpha_k} \Psi_k(\xi)$$

$$\begin{aligned} &\mu(x; \lambda, \alpha) \\ &\sigma^2(x; \lambda, \alpha) \end{aligned}$$

Key technical ingredients:

- Non-intrusive spectral projection (NISP) for uncertainty propagation
- Simultaneous Bayesian inference of λ and α via Markov chain Monte Carlo
- Variance decomposition of prediction uncertainty

Prediction mean

$$\mu(x) = \mathbb{E}_{\lambda, \alpha}[\mu(x; \lambda, \alpha)]$$

Prediction variance

$$\sigma^2(x) = \mathbb{V}_{\lambda, \alpha}[\mu(x; \lambda, \alpha)] + \mathbb{E}_{\lambda, \alpha}[\sigma^2(x; \lambda, \alpha)]$$

Posterior

Model error

Polynomial chaos (PC) expansions are well-suited for uncertainty propagation and attribution

$$f\left(x; \lambda + \sum_k \overset{\text{PC}}{\alpha_k} \Psi_k(\xi)\right) \overset{\text{NISP}}{\simeq} \sum_k \overset{\text{physical}}{f_k(x; \lambda)} \overset{\text{model-error}}{\alpha_k} \Psi_k(\xi)$$

$\mu(x; \lambda, \alpha)$
 $\sigma^2(x; \lambda, \alpha)$

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Posterior

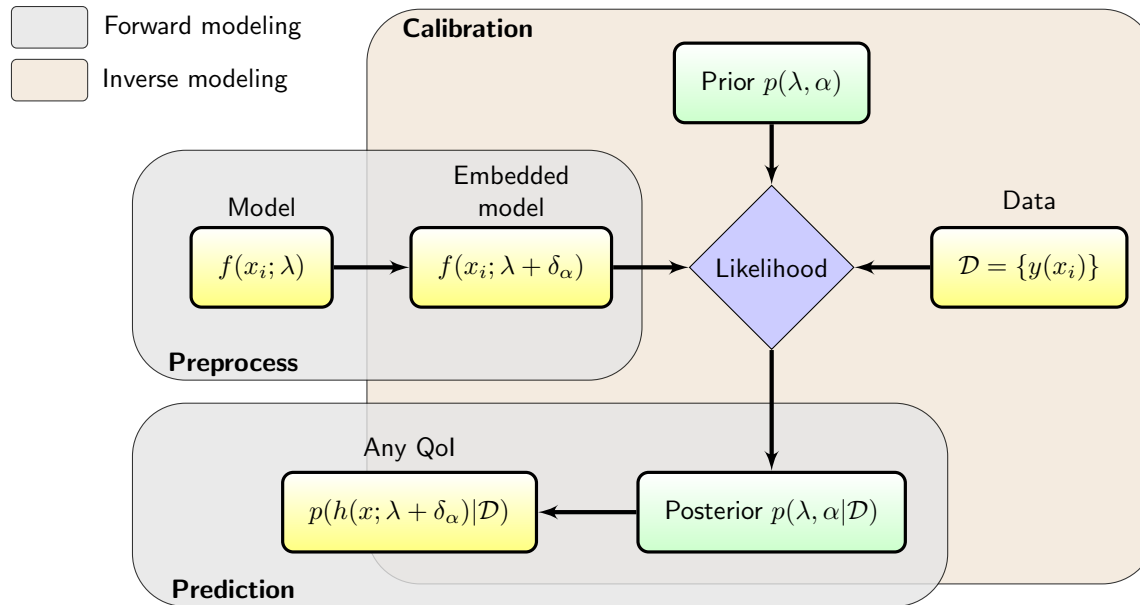
Model error

Does not shrink
with more data

Embedded, non-intrusive workflow is implemented in UQTK



Funded by FASTMath SciDAC Institute, www.sandia.gov/uqtoolkit



Applications

- Earth System Land Model (BER)
- Chemical kinetic modeling (BES)
- Plasma surface interactions (FES)
- Turbulence modeling (DARPA)

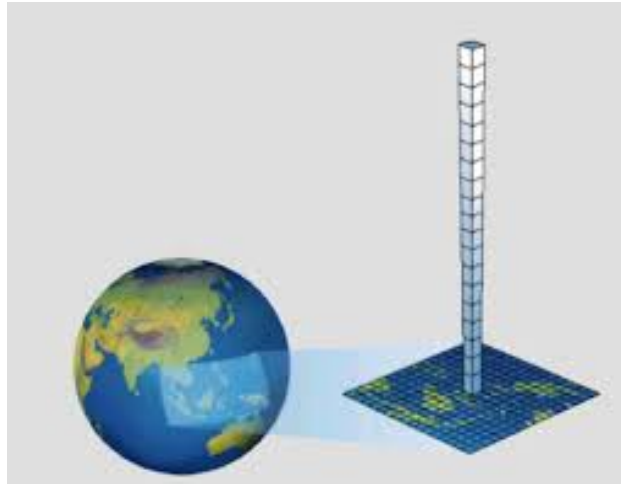
Method: Sargsyan, Najm, Ghanem, IJCK (2015); Sargsyan, Huan, Najm, IUQ (2019).

Applications: Huan et. al, AIAA J (2018); Hakim et. al, CTM (2018); Cekmer et. al, IUQ (2018); Rizzi et. al, CMAME (2019).

Major application of focus: Earth System Land Model

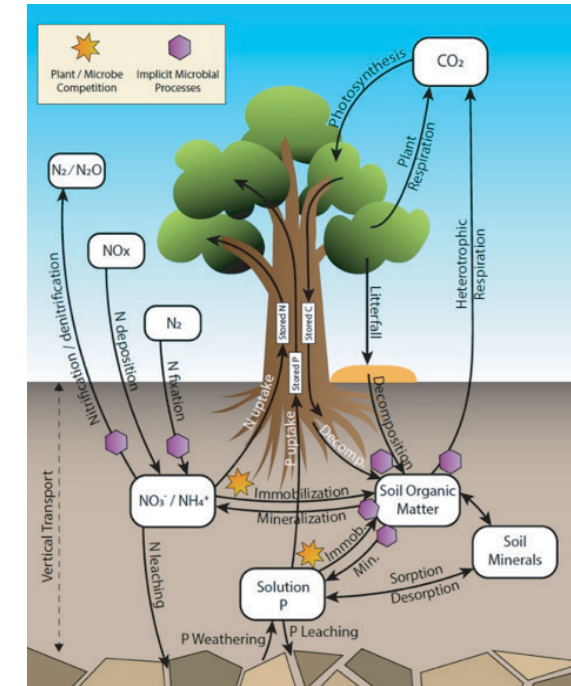
funded by **BER+ASCR**

- US DOE sponsored Earth system model
- Land, atmosphere, ocean, ice, human system components
- High-resolution, employ DOE leadership-class computing facilities



National Energy Research
Scientific Computing Center

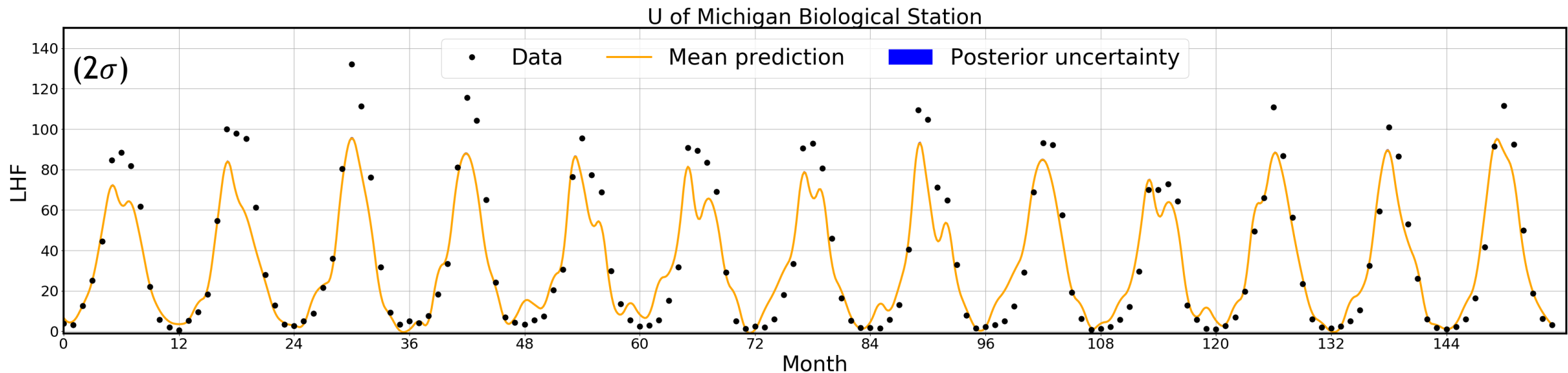
Serving as a UQ Lead for Land Model:
Direct UQ impact on several land components,
collaboration with multiple NLs



Calibration without embedded model error



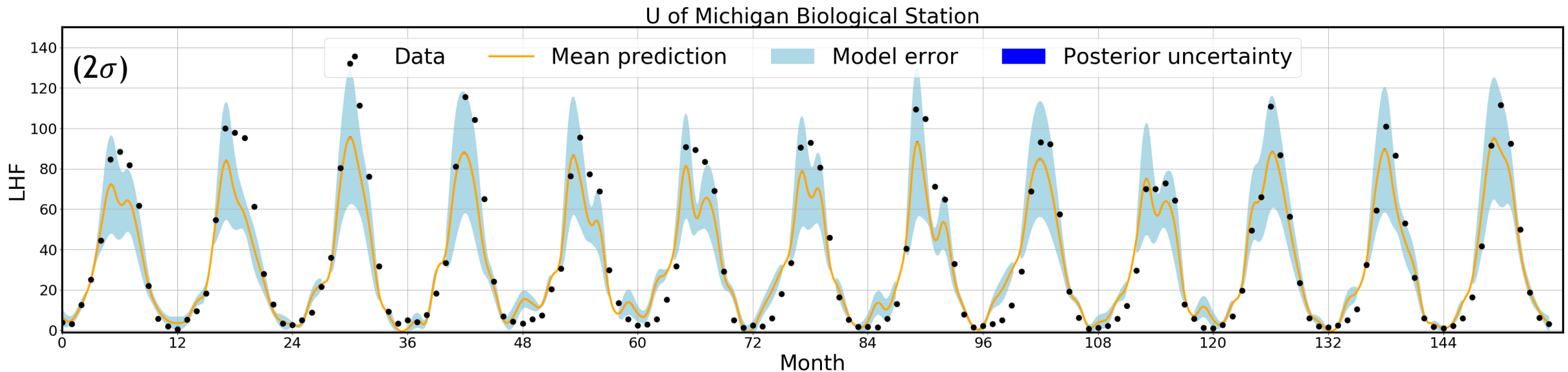
Land model calibration given FLUXNET observations



- Summer month peaks are not captured
- Posterior uncertainty negligible

LHF = Latent Heat Flux

Land model calibration given FLUXNET observations

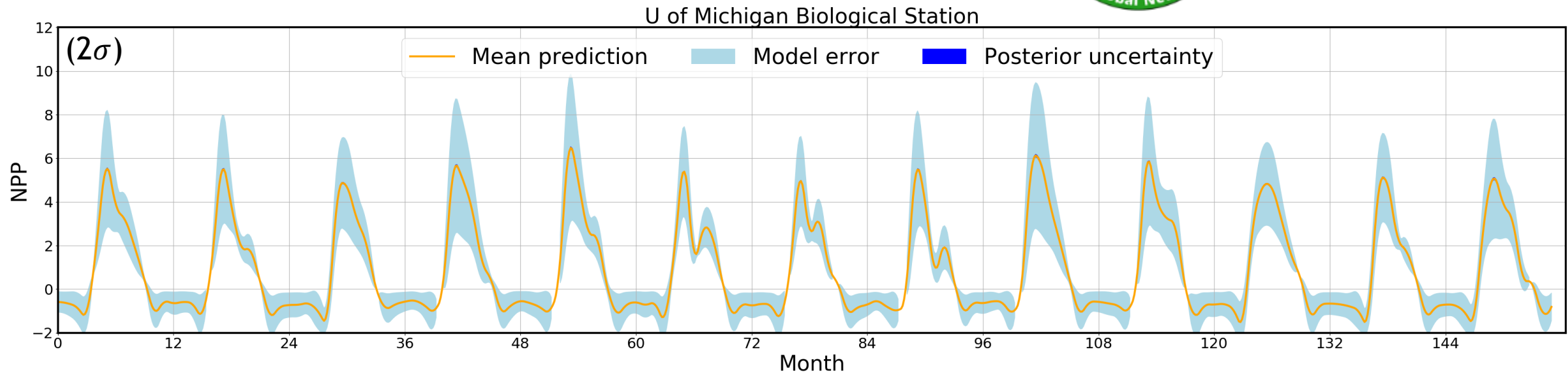


- Model error component dominates
- Captures model deficiency in summer months
- Indicates model improvement opportunities
- For further improvement: more intrusive embedding

LHF = Latent Heat Flux



Land model calibration given FLUXNET observations

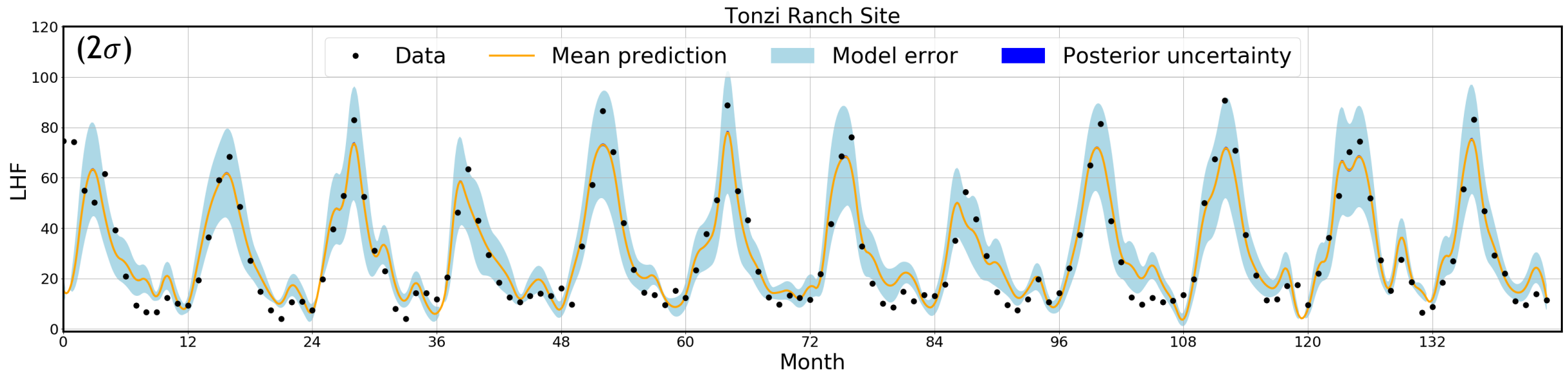


- Can be piped to human component or atmosphere model as a boundary condition

NPP = Net Primary Productivity



Land model calibration given FLUXNET observations



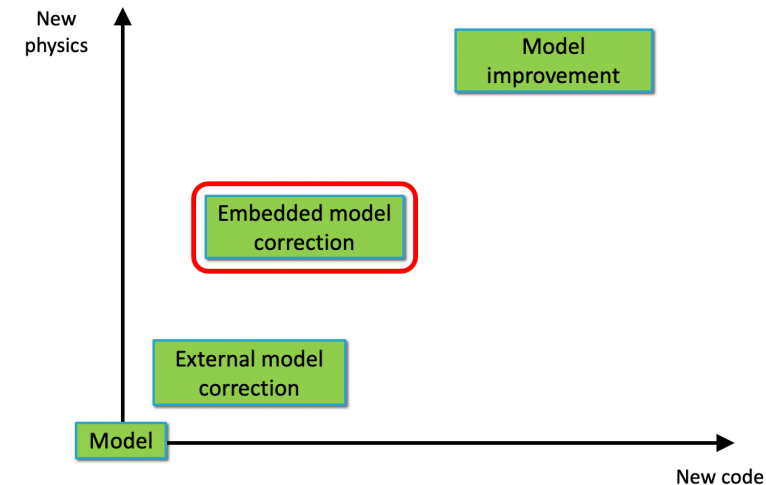
- Assumption: model goes wrong in a similar way

Unique capability for model exploration via **embedded** model error representation

Non-intrusive

$$y(x_i) = f(x_i; \lambda + \delta(x_i)) + \epsilon_i$$

- Allows meaningful extrapolation
 - Respects physics
 - Disambiguates model and data errors
- Predictive uncertainty attribution
- surrogate errors
 - data noise
 - parametric uncertainty
 - **structural errors**



- Code available via UQTK (www.sandia.gov/uqtoolkit)
- Impacted many programs DOE/DOD/SNL
- Applied outside immediate group
- Provides alternative for external correction approach currently available in DAKOTA

UQTK

Prediction variance

=

parametric uncertainty

+

data noise

+

model error



Current/Future

- Method development under FASTMath and BER/ASCR SciDAC partnership
 - challenging Bayesian problem: approximate likelihood methods, informative priors
 - parameter *field* embedding: proven need for climate land model
 - probabilistic neural networks: many technical analogies; *see Ahmad Rushdi's poster*

Key References

- K. Sargsyan, H. Najm, R. Ghanem, “On the Statistical Calibration of Physical Models”, Int. J. Chem. Kinetics, 47(4), 246-276, 2015, ~30 citations.
- K. Sargsyan, X. Huan, H. Najm. “Embedded Model Error Representation for Bayesian Model Calibration”, arXiv:1801.06768, in press, Int. J. Uncert. Quant., 2019.



Additional slides

Model error can be defined in a variety of ways



In this work, model error is the difference between our model and the 'truth' model behind noisy data


$$\begin{array}{ccccc} f(x_i; \lambda) & \text{vs} & y(x_i) = g(x_i) + \epsilon_i \\ \text{model} & & \text{data} & \text{truth} & \text{noise} \end{array}$$

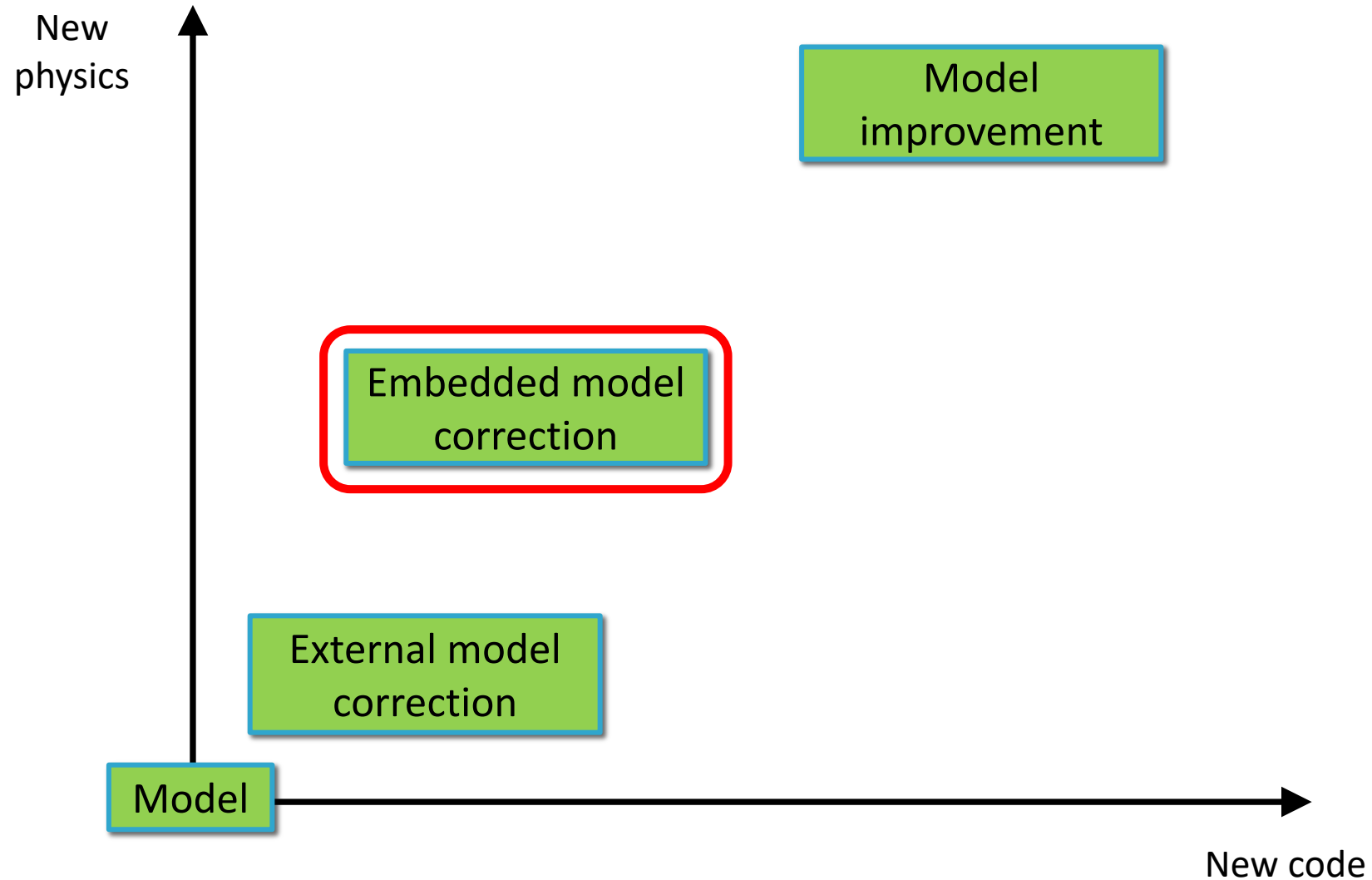
Model error is associated with

- Simplifying assumptions, parameterizations
- Mathematical formulation, theoretical framework

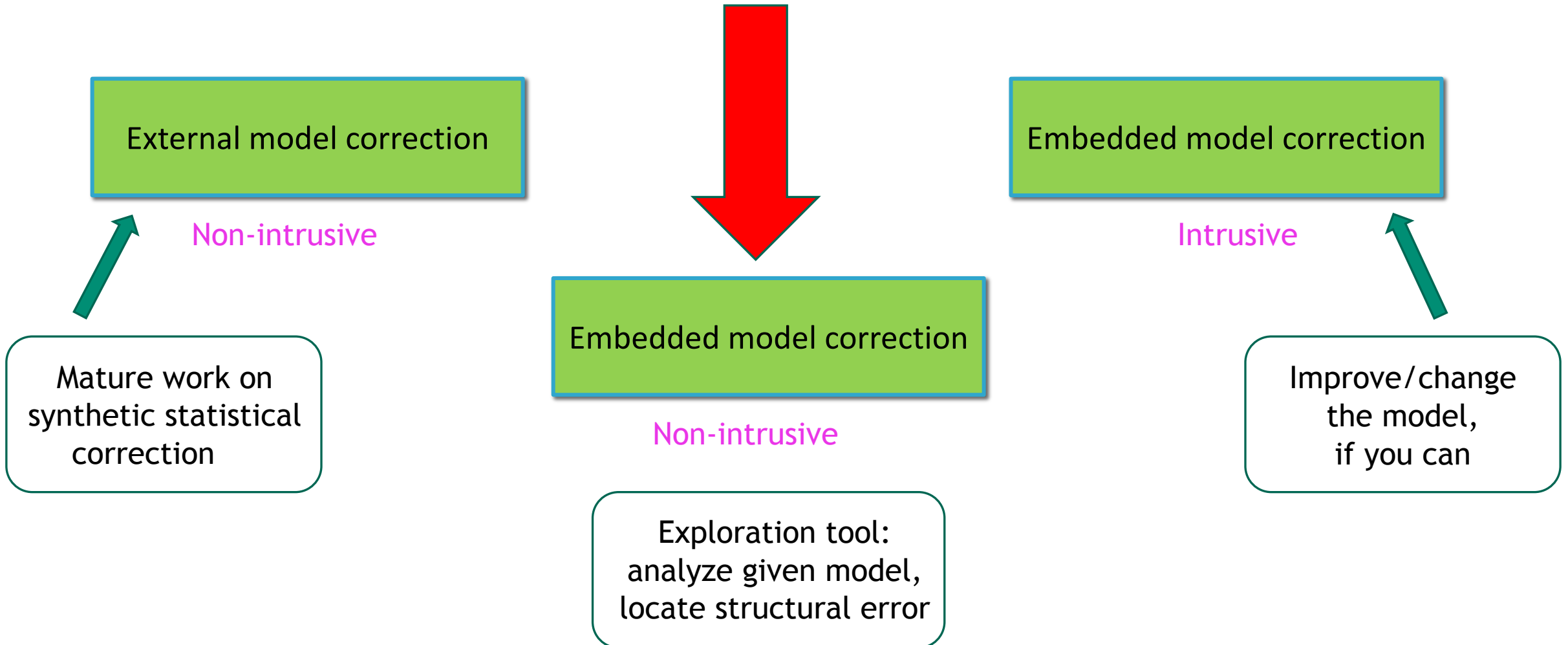
Very loaded concept
... otherwise called
(with altered meanings)

- model discrepancy
- model structural error
- model inadequacy
- model misspecification
- model form error
- model uncertainty

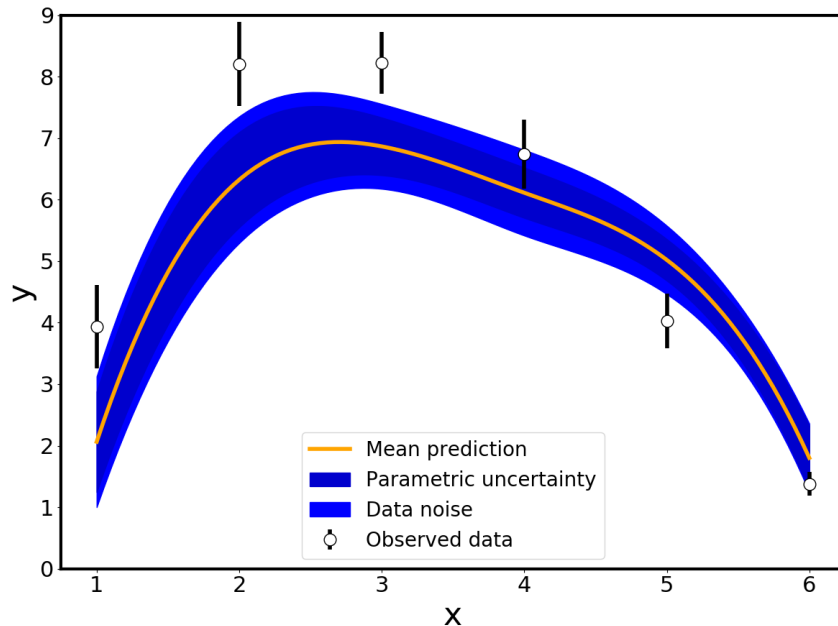
There is mature work in *statistical* model corrections, without physical interpretation



We develop approach for model structural error representation: **embedded, but non-intrusive**



Prediction variance accounts for parametric uncertainty and data noise



$$f(x; \lambda) \approx y(x)$$

This is business as usual....

Posterior uncertainty (PU)

parametric uncertainty

+

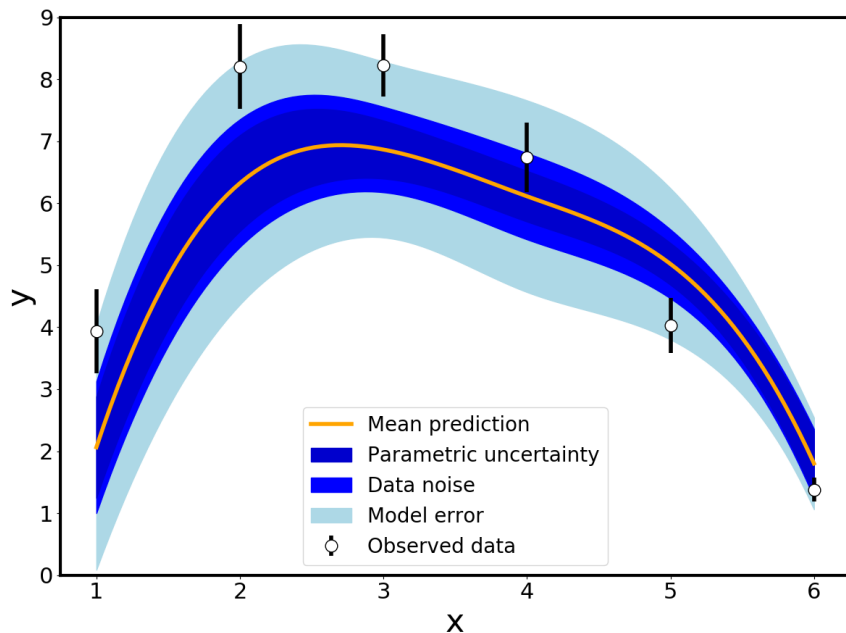
data noise

... assuming the model is perfect, i.e.
represents the truth behind data

Elephant in the room:



model *structural* error



Model

Data

$$f(x; \lambda) \approx y(x)$$

Uncertainty decomposition of model prediction
needs to account for model error

Posterior uncertainty (PU)

parametric uncertainty

+

data noise

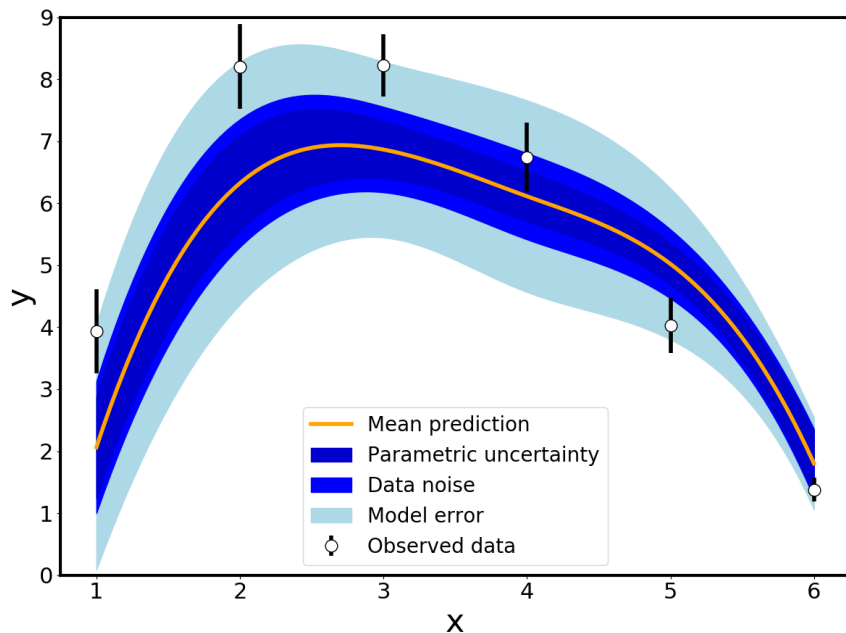
+

+

model error

ME

We want to incorporate model structural error in predictions



Model

Data

$$f(x; \lambda) \approx y(x)$$

Uncertainty decomposition of model prediction needs to account for model error

Posterior uncertainty (PU)

parametric uncertainty

+

data noise

+

+

model error

ME

External correction is not satisfactory for physical models



$$y_i = \underbrace{f(x_i; \lambda) + \delta(x_i)}_{\text{truth } g(x_i)} + \epsilon_i$$

- Explicit additive statistical model for model error [KOH, 2001]
- Potential violation of physical constraints
- Disambiguation of model error $\delta(x_i)$ and data error ϵ_i
- Yes, priors help: [Brynjarsdottir and O'Hagan, 2014], [Plumlee, 2017]
- Calibration of model error on measured observable does not impact the quality of model predictions on other QoIs
- Physical scientists are unlikely to augment their model with a statistical model error term on select outputs
 - Calibrated predictive model: $f(x; \lambda) + \delta(x)$ or $f(x; \lambda)$?
- Problem is highlighted in model-to-model calibration ($\epsilon_i = 0$)
 - no a priori knowledge of the statistical structure of $\delta(x)$

Calibrate $f(x; \lambda)$, given data $g(x)$

x are operating conditions, design parameters, various QoIs

λ are model parameters to be inferred/calibrated

- **Default:** Ignore model errors:

$$g(x) = f(x; \lambda) + \epsilon$$

- Biased or overconfident physical parameters
- Wrong model predictions

- **Conventional:** Correct for model errors:

$$g(x) = f(x; \lambda) + \delta(x) + \epsilon$$

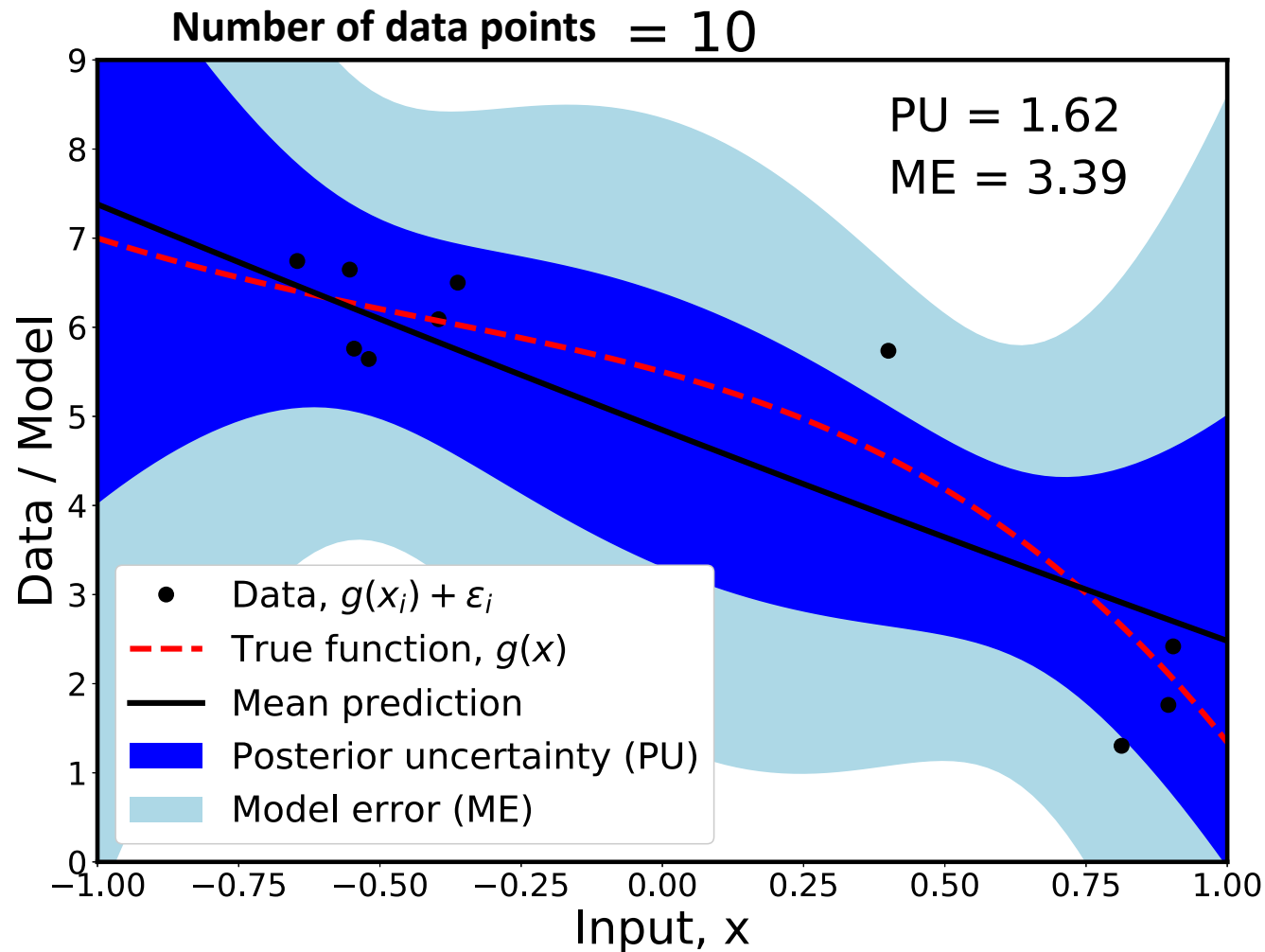
- Physical parameters are ok
- Wrong model predictions (data-specific corrections)
- Model and data errors mixed up

- **What we do:** Correct *inside* the model:

$$g(x) = f(x; \lambda + \delta(x)) + \epsilon$$

- Embedded model error
- Preserves model structure and physical constraints
- Disambiguates model and data errors
- Allows meaningful extrapolation

With more data points, posterior uncertainty (PU) goes down, but model error (ME) saturates to a limiting value



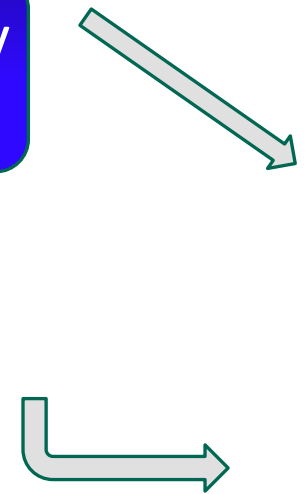
Predictive uncertainty

=

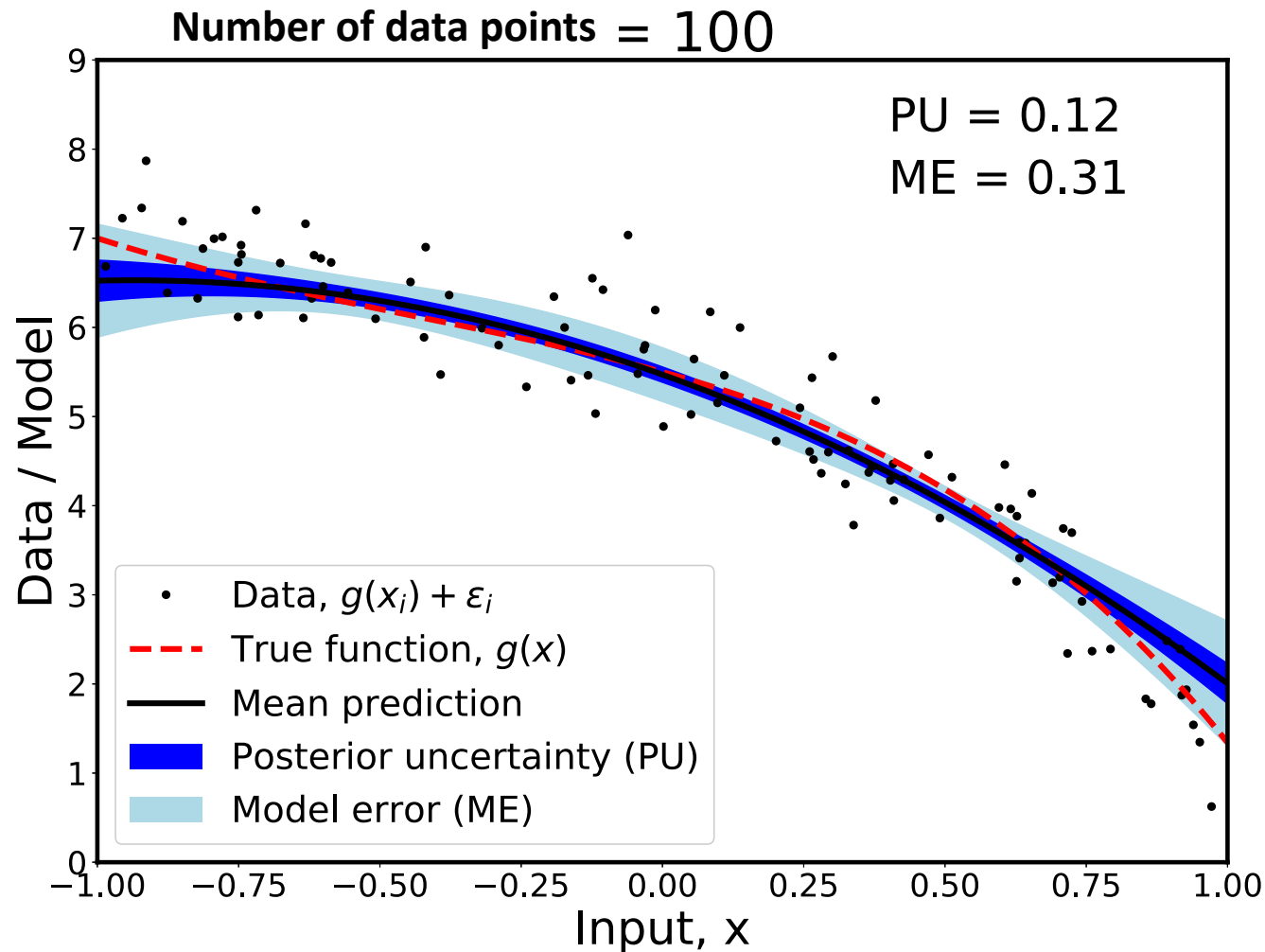
Post. uncertainty
(PU)

+

Model error
(ME)



With more data points, posterior uncertainty (PU) goes down, but model error (ME) saturates to a limiting value



Predictive uncertainty

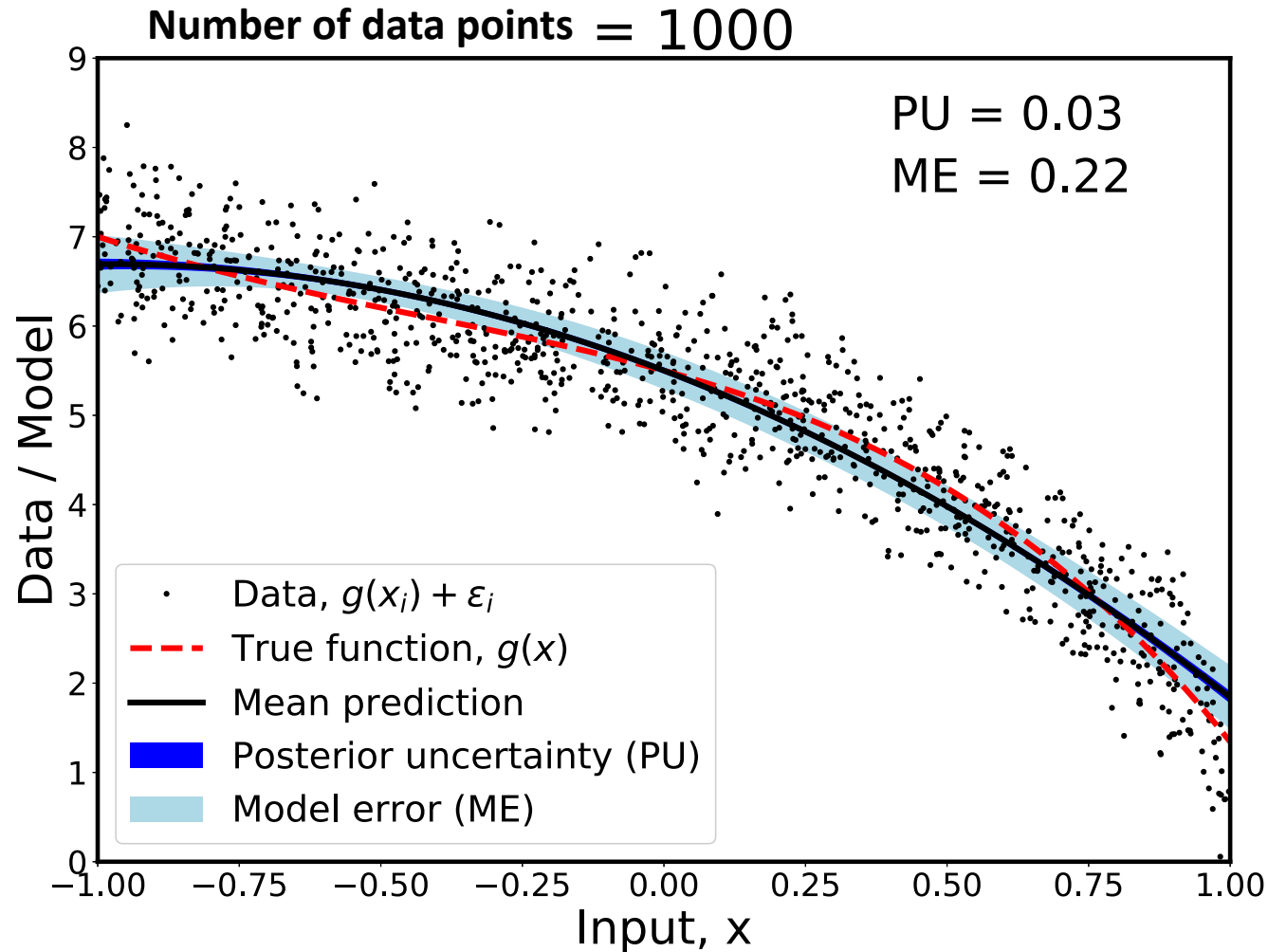
=

Post. uncertainty
(PU)

+

Model error
(ME)

With more data points, posterior uncertainty (PU) goes down, but model error (ME) saturates to a limiting value



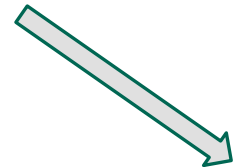
Predictive uncertainty

=

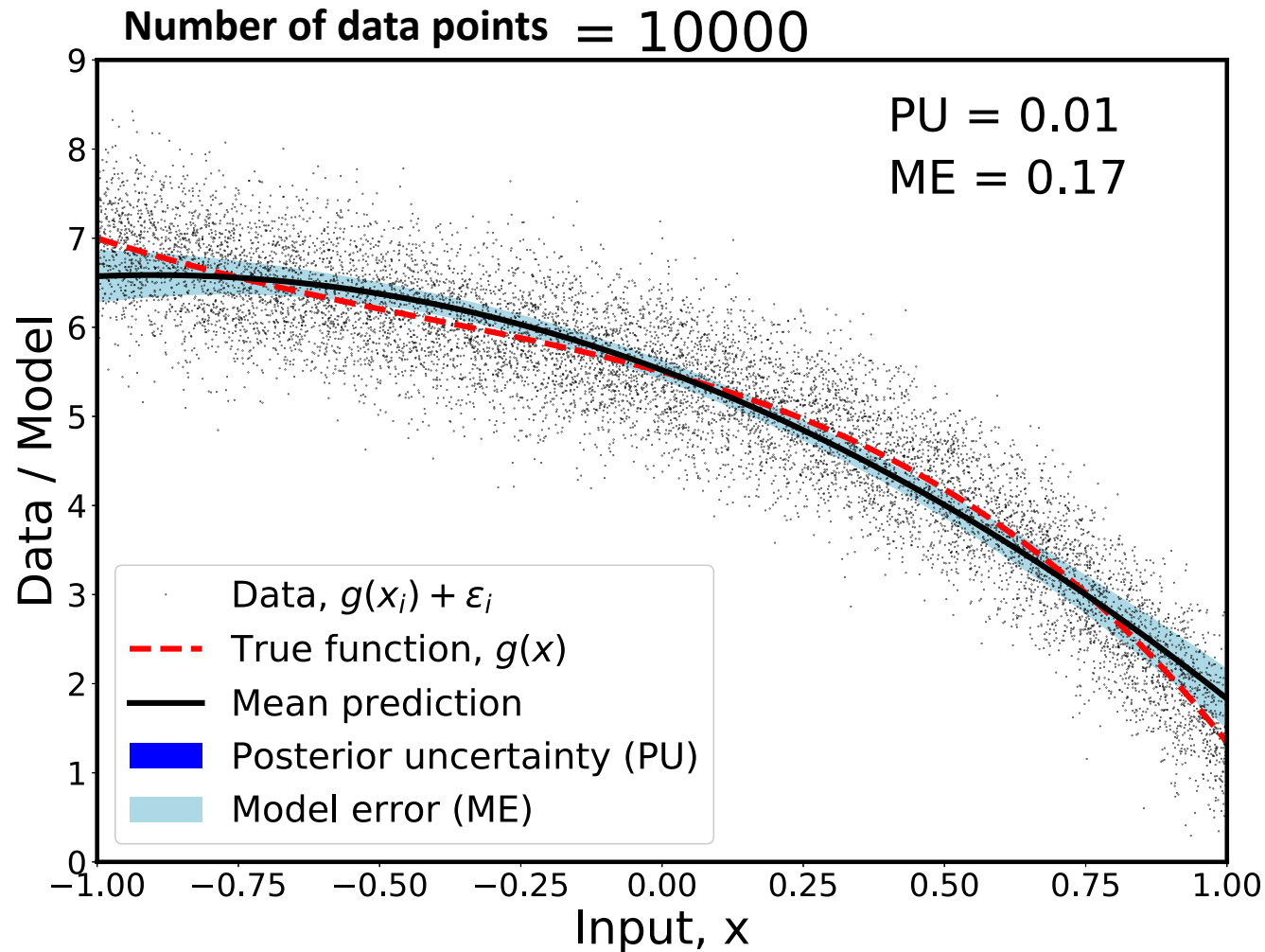
Post. uncertainty
(PU)

+

Model error
(ME)



Irrespective of data amount, model error captures the discrepancy between model and the truth



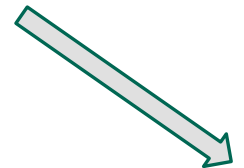
Predictive uncertainty

=

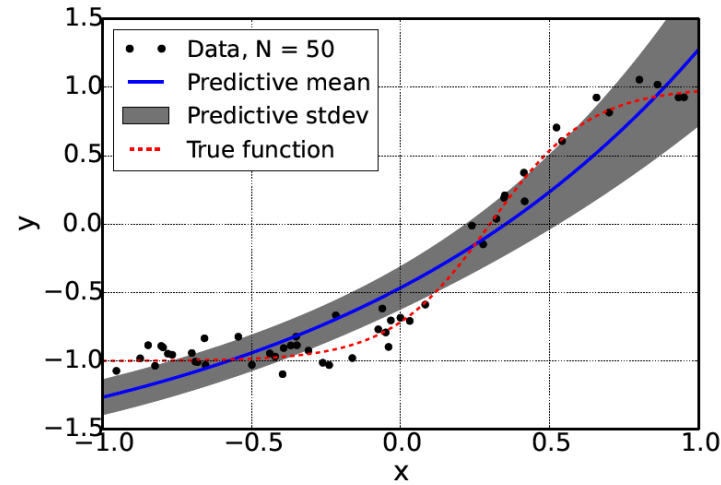
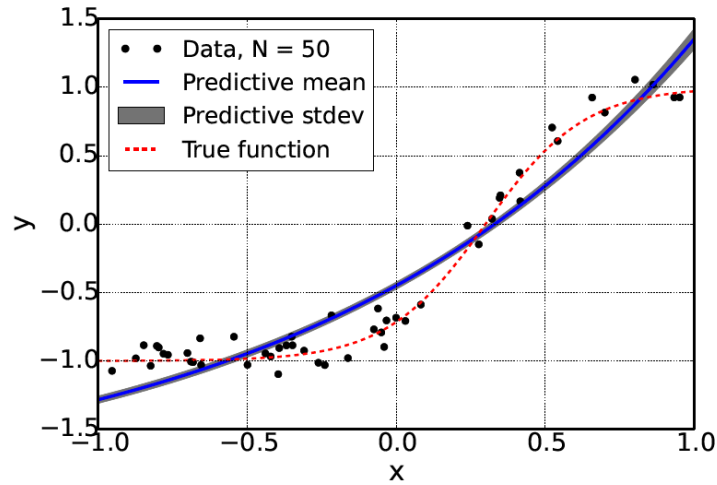
Post. uncertainty
(PU)

+

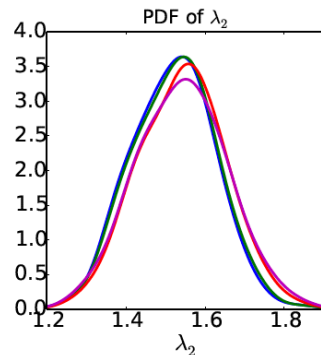
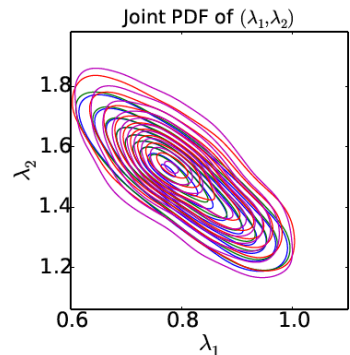
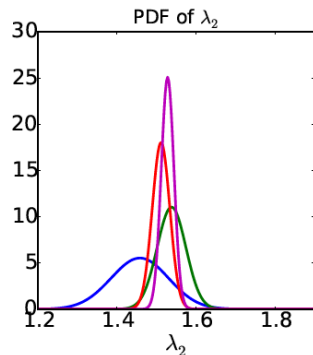
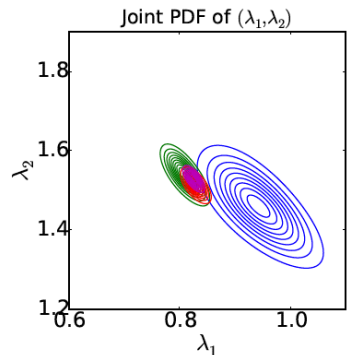
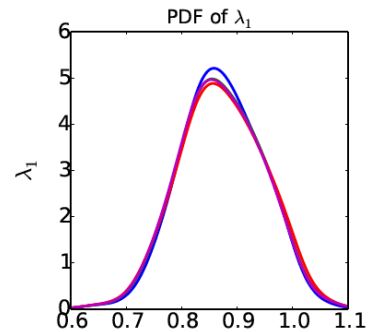
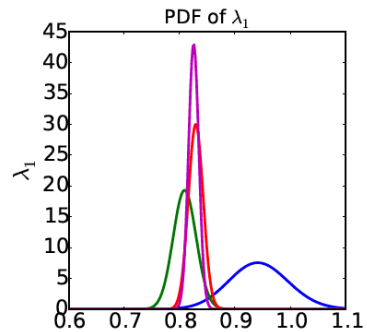
Model error
(ME)



Back to toy example

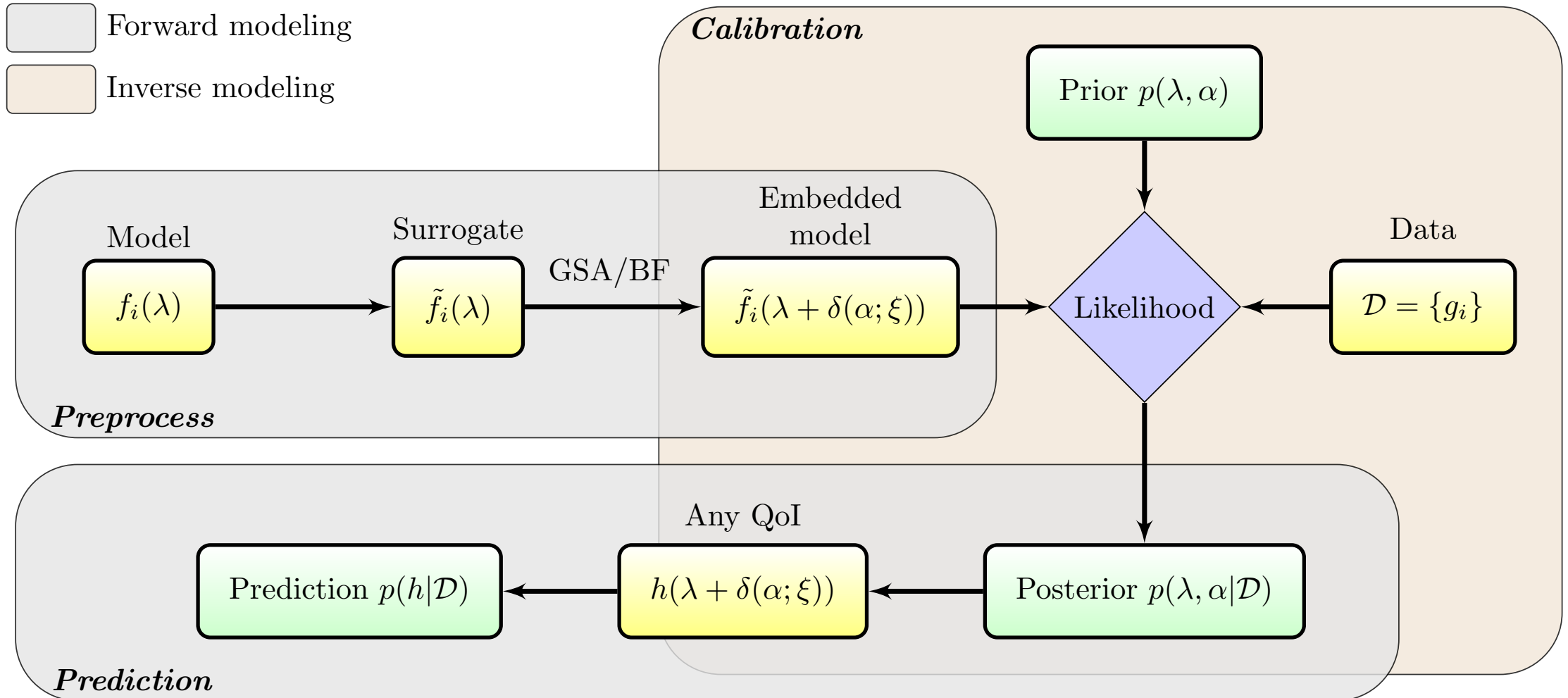


Predictive uncertainty captures model error



Stable prediction of “physical” parameters of the exponential function

Workflow



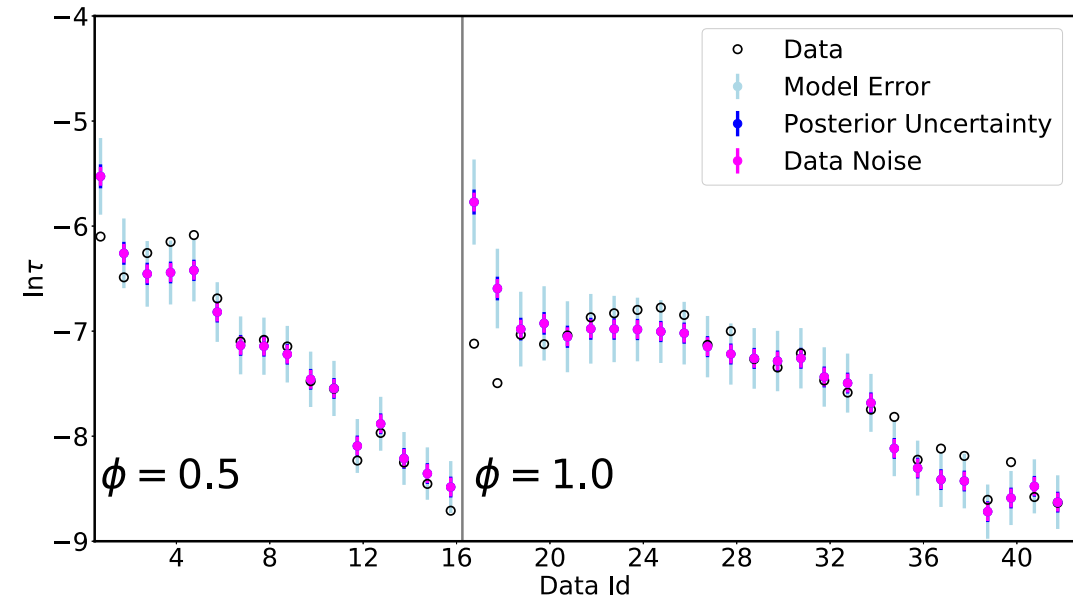
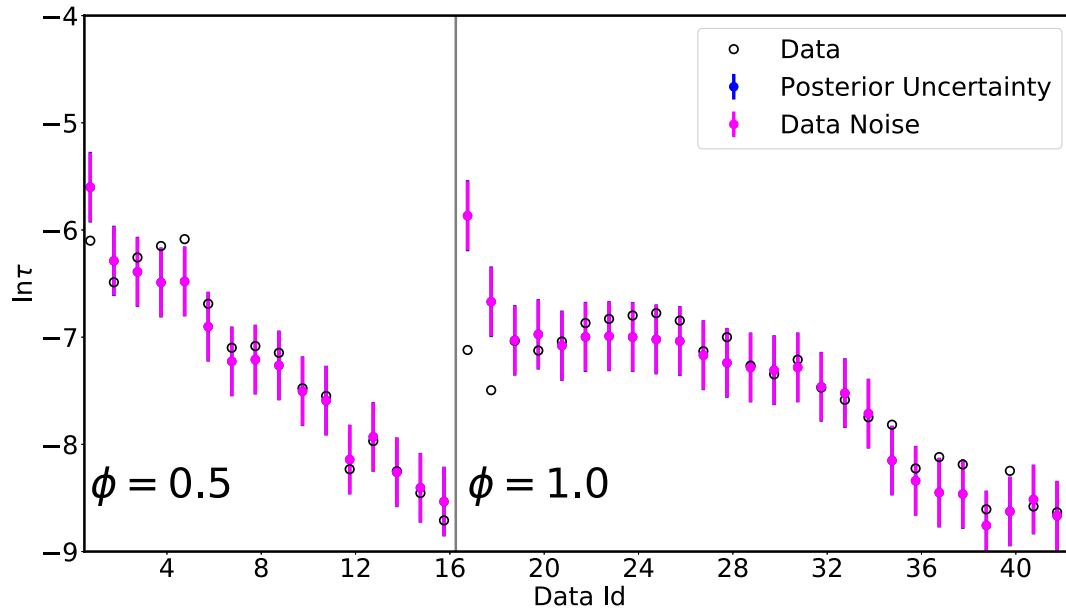
Application 2: Chemistry

funded by BES

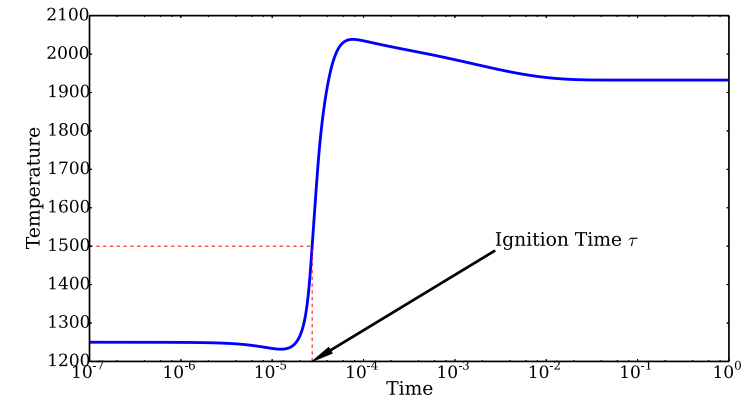


- A sandbox for developing the method
- Calibrating a simple 2-step reaction mechanism given high-fi model or experimental data

Without model error,
all the discrepancy
is attributed to data noise



Qol: log-ignition time



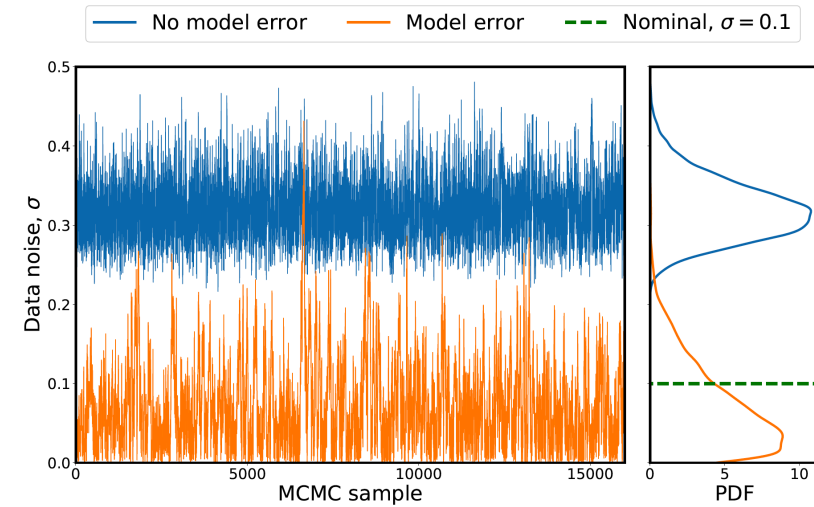
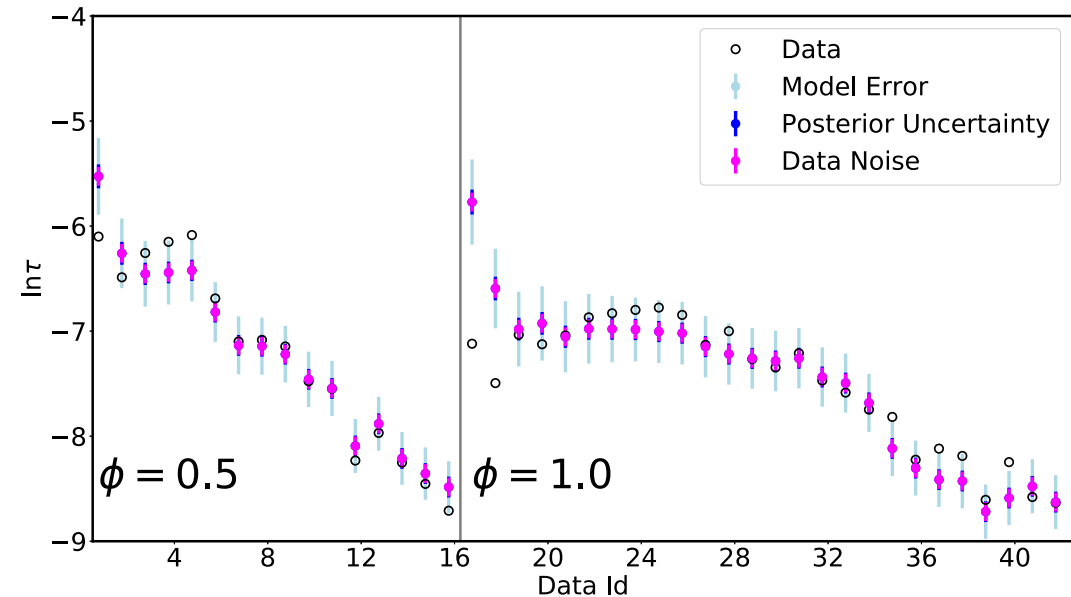
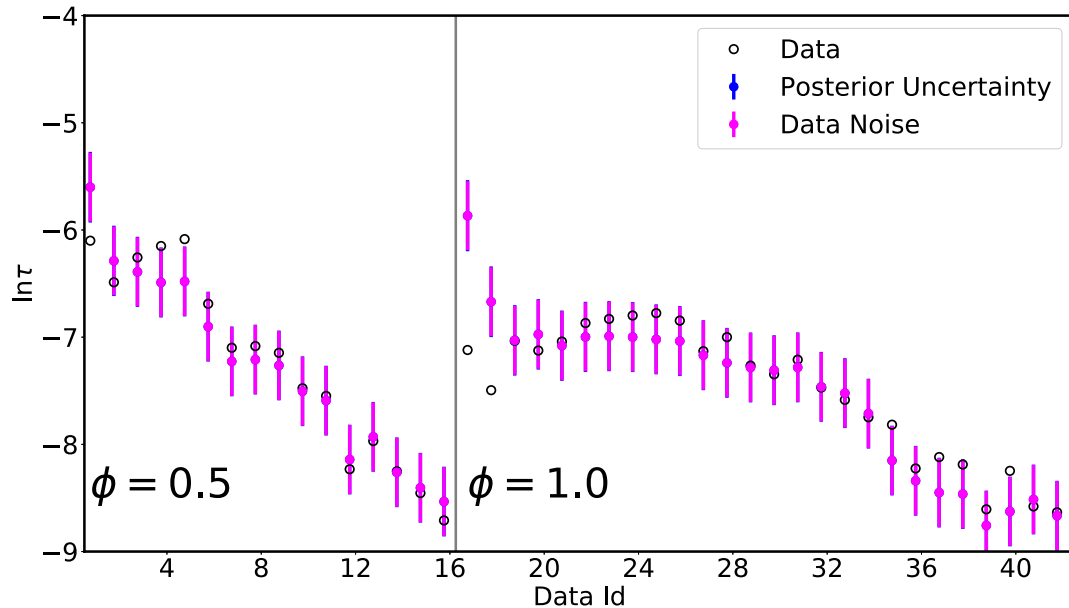
Application 2: Chemistry

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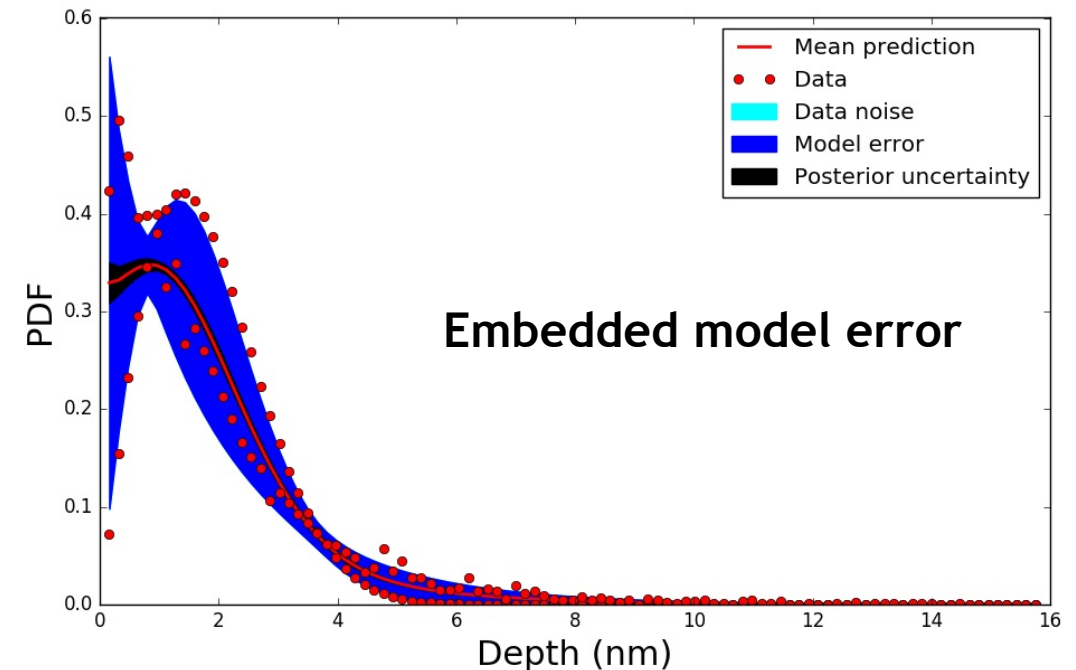
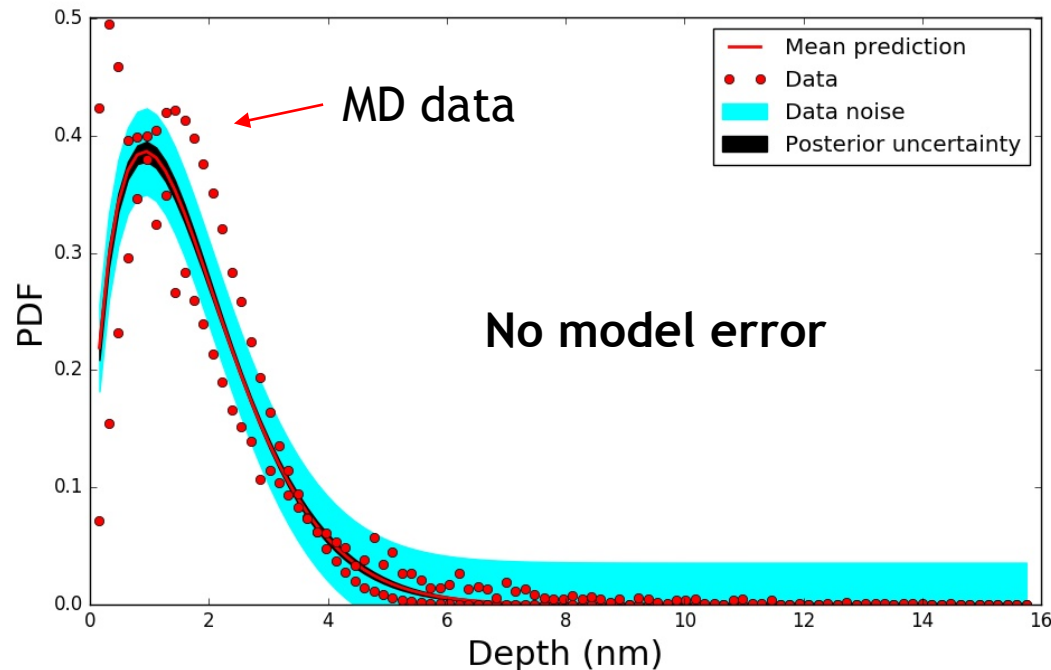


Application 3: Plasma Surface Interaction funded by FES+ASCR



- Multi-institution partnership, but direct collaboration with ORNL and UTK
- Constructing uncertain input profiles for tungsten depth to propagate through Xolotl (PSI code)

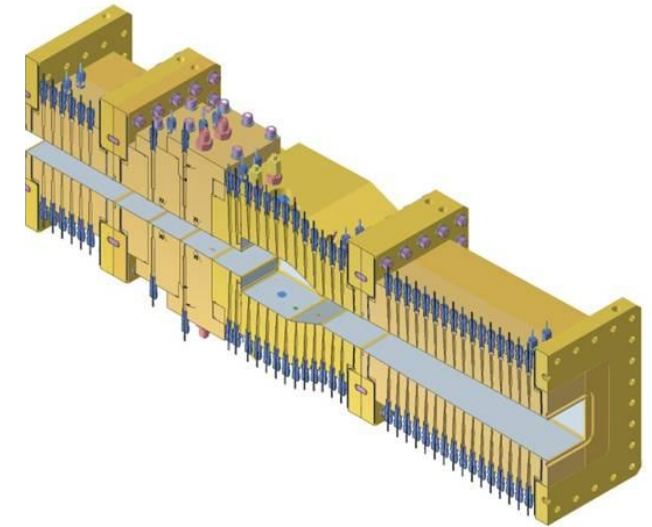
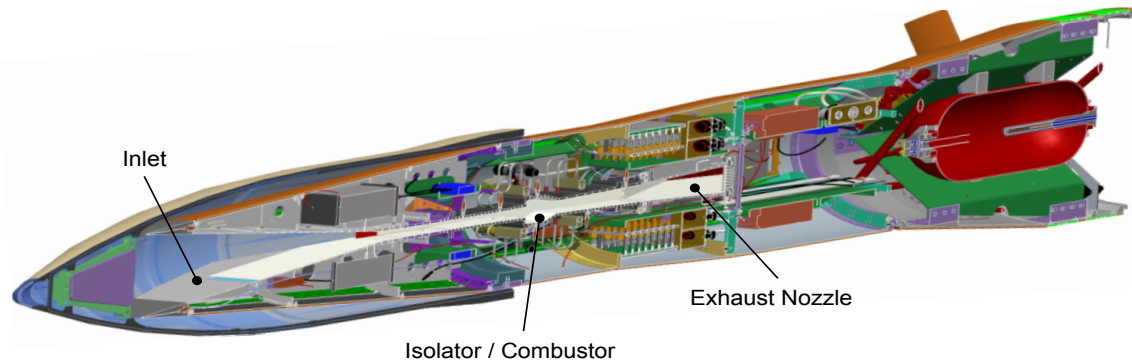
Ignoring model error wrongly attributes uncertainty to data



O. Cekmer, K. Sargsyan, S. Blondel, H. Najm, D. Bernholdt, B.D. Wirth, "Uncertainty quantification for incident helium flux in plasma-exposed tungsten", *Int. J. Uncertainty Quantification*, Vol. 8, No. 5, p.429–446, 2018.

Application 4: Turbulent Flow

funded by DARPA



Advance the state of the art in UQ methods and software, in Large Eddy Simulation (LES) of a laboratory scale Scramjet combustor
NASA Langley Hypersonic International Flight Research and Experimentation (HIFiRE) configuration

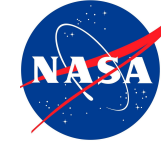
Application 4: Turbulent Flow

funded by DARPA

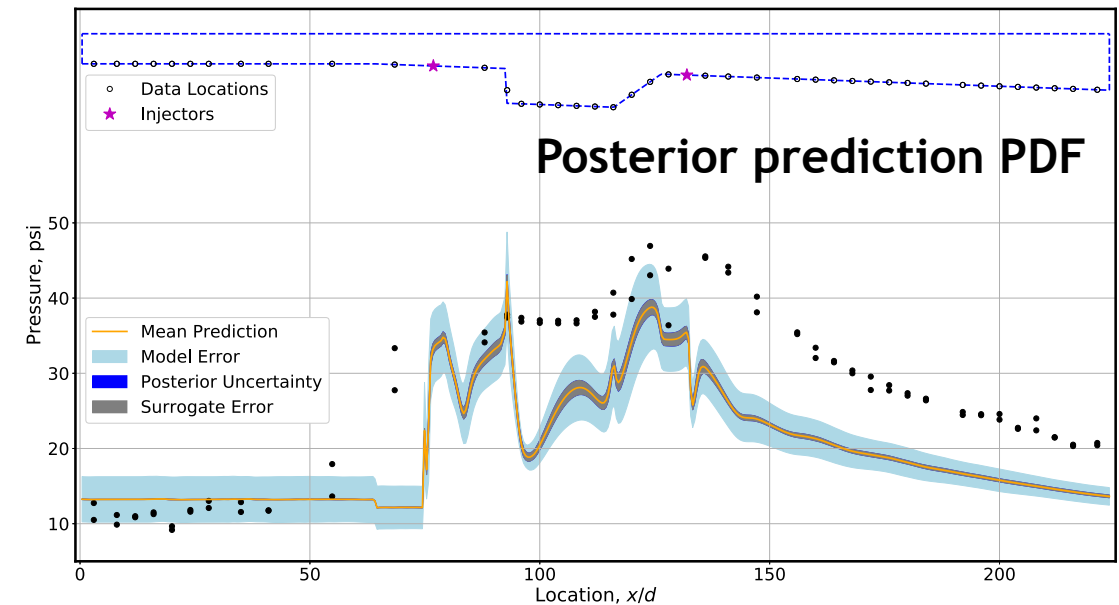
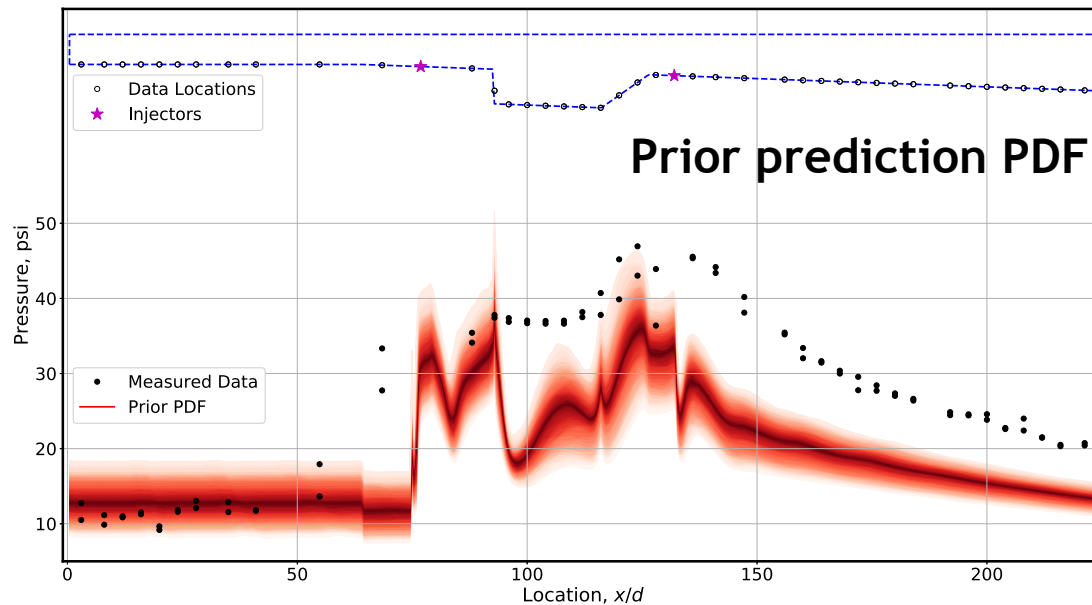


- Major UQ challenges for turbulent flow (LES)
 - Nonlinear dynamics
 - Large number of uncertain parameters
 - **LES model structural error**
 - Optimize design under uncertainty

Experimental data
obtained from NASA



Model error is the main contributor
of the predictive variance



X. Huan, C. Safta, K. Sargsyan, G. Geraci, Michael S. Eldred, Zachary P. Vane, G. Lacaze, Joseph C. Oefelein, Habib N. Najm, "Global Sensitivity Analysis and Estimation of Model Error, toward Uncertainty Quantification in Scramjet Computations", *AIAA Journal*, Vol. 56, No. 3, p.1170–1184, 2018

Applications (other)



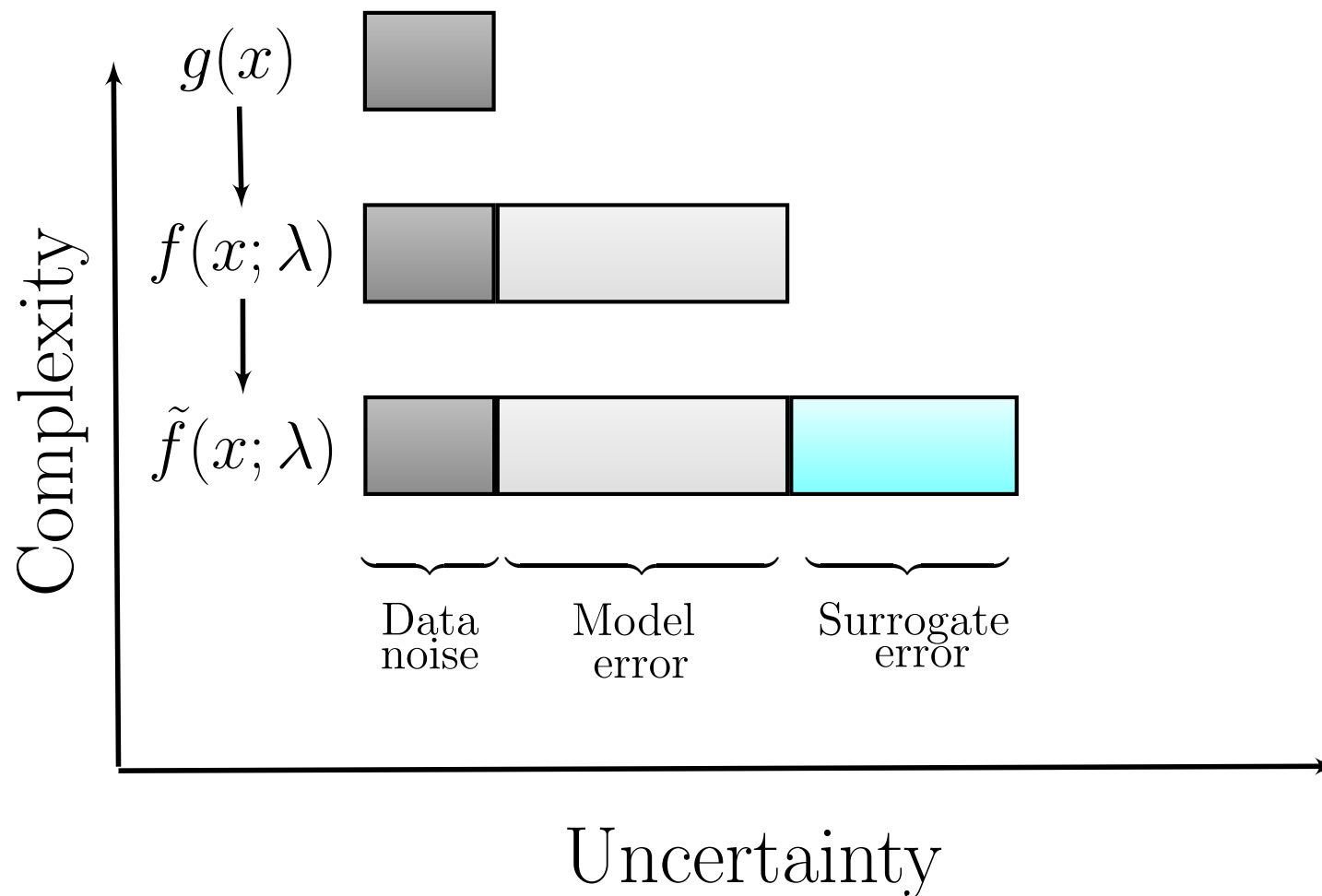
Thermodynamics (DOE EERE), work by Bert Debusschere

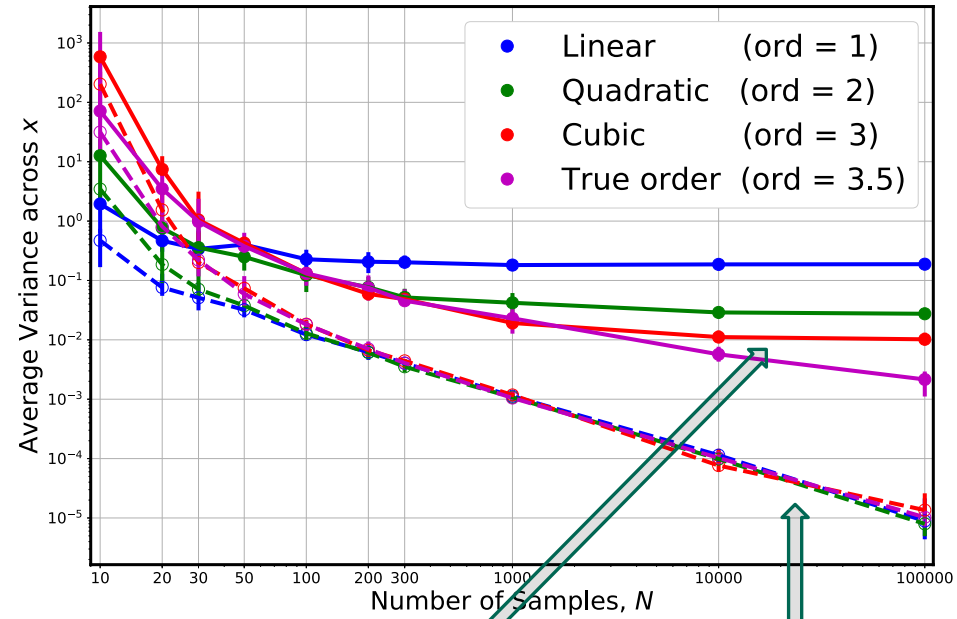
- The UQTk model error tools have been used to study the impact of using lower fidelity models for the thermodynamics of redox active materials that are used in water splitting methods for Hydrogen generation.
- This work is part of the HydroGEN Advanced Water Splitting Materials project (<https://www.h2awsm.org/>). HydroGEN is funded by DOE's Fuel Cell Technologies Office in the Office of Energy Efficiency and Renewable Energy (EERE).

Flood Modeling (NSF/UMichigan), collaboration/advising within DOE SCGSR program

Materials Science (DOE/LDRD): work by Reese Jones, Francesco Rizzi

Uncertainty Budget

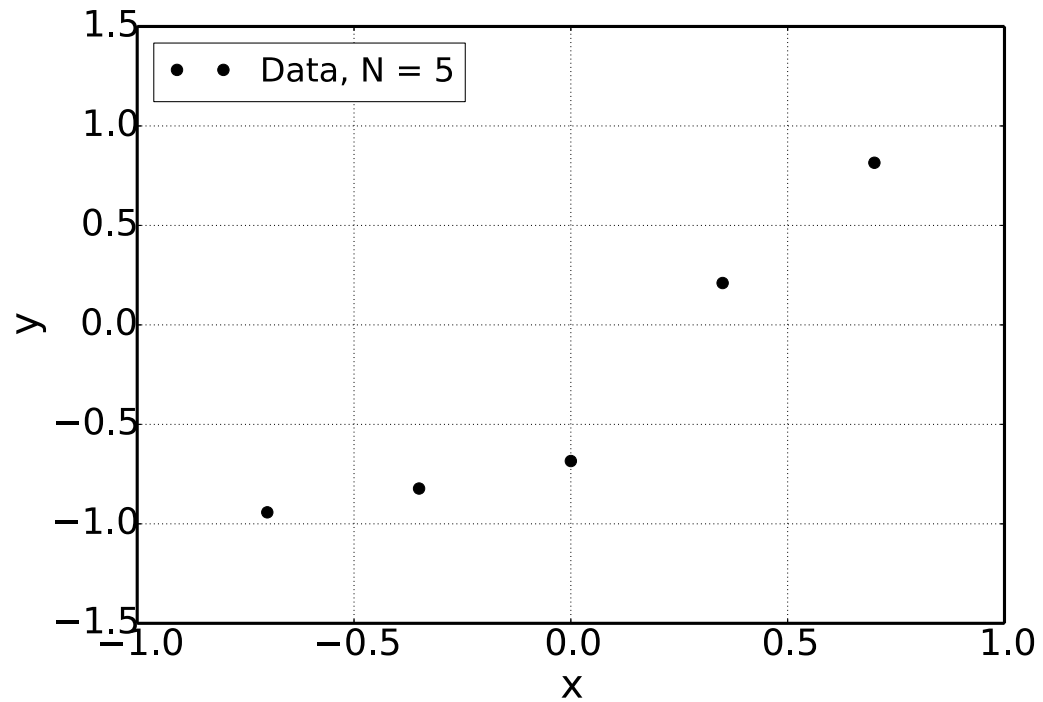




Model error
(ME)

Post. uncertainty
(PU)

Wrong model leads to biased estimation

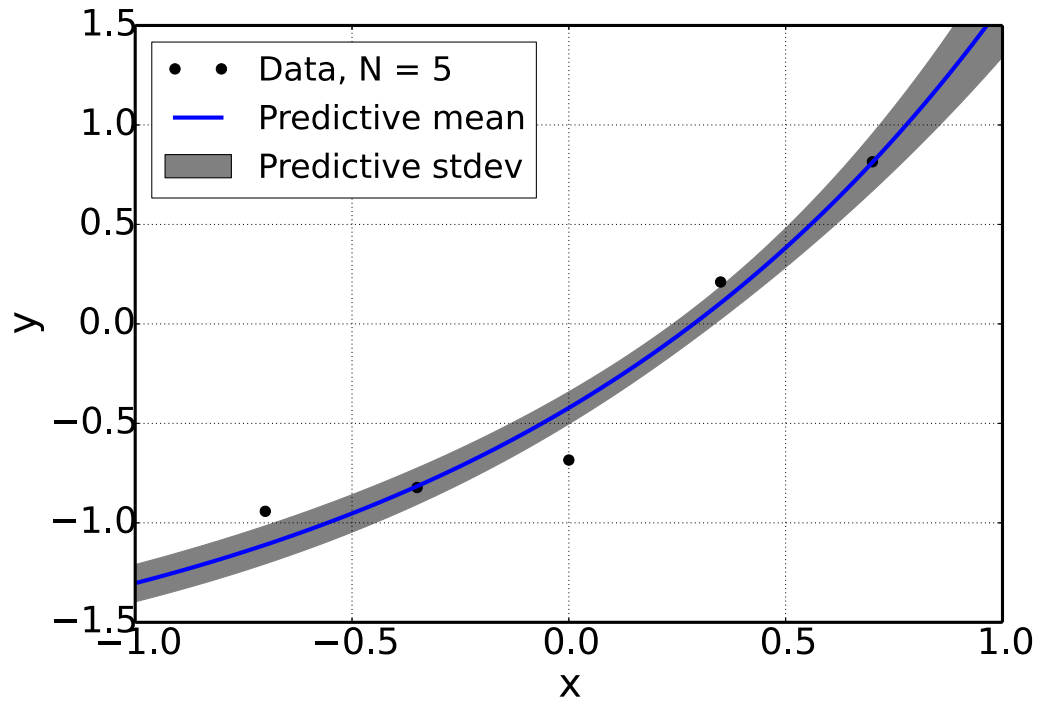


Given noisy data

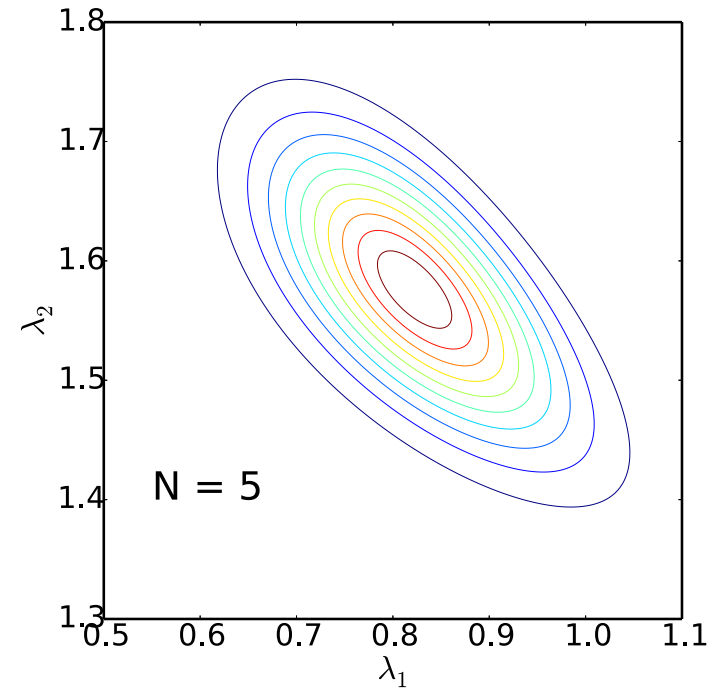
Wrong model leads to biased estimation



Model prediction vs data



Posterior PDF of exponential parameters

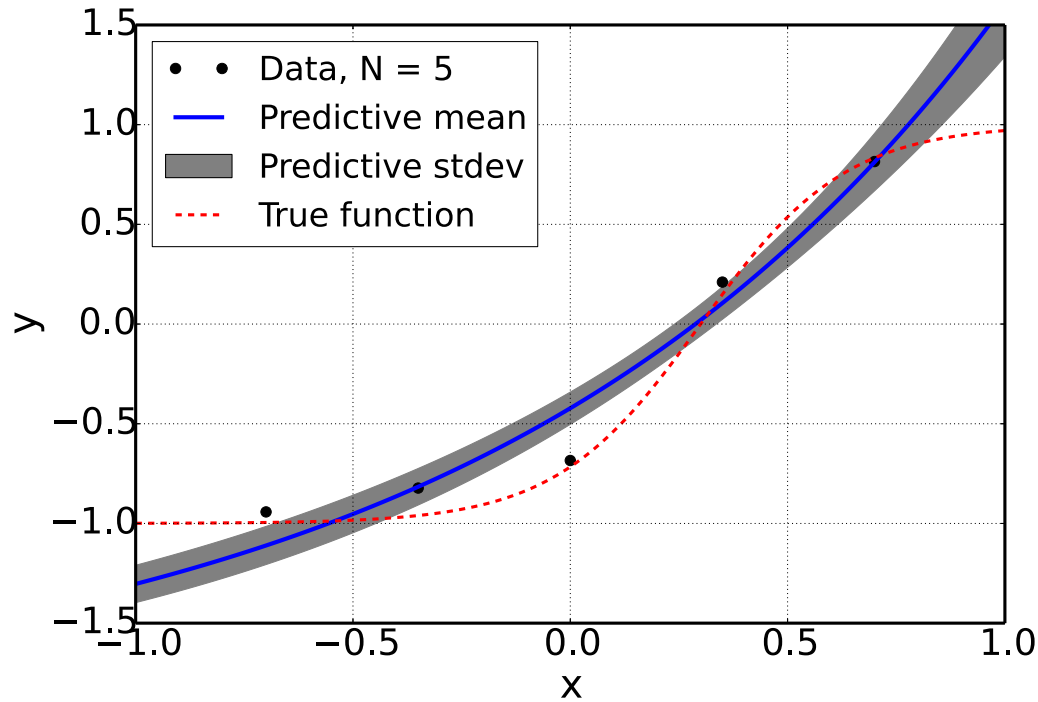


Calibrate an exponential model

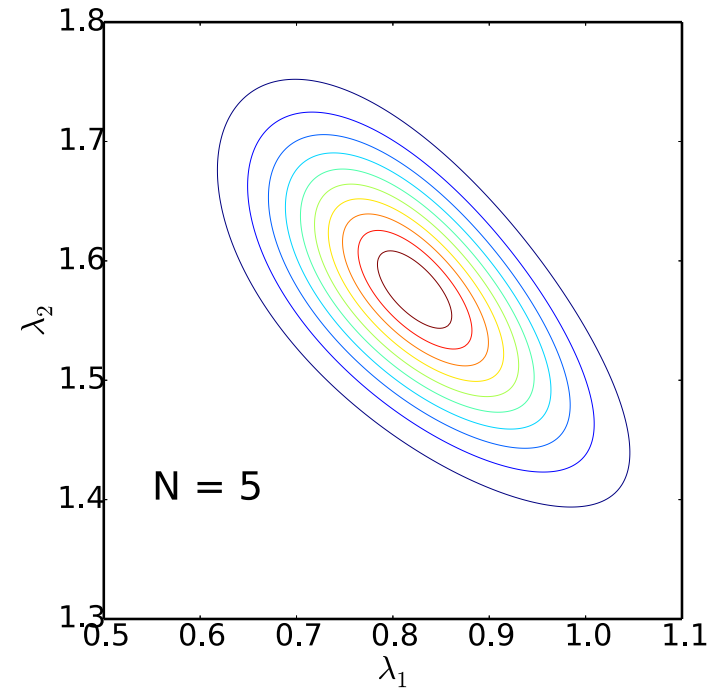
Wrong model leads to biased estimation



Model prediction vs data



Posterior PDF of exponential parameters

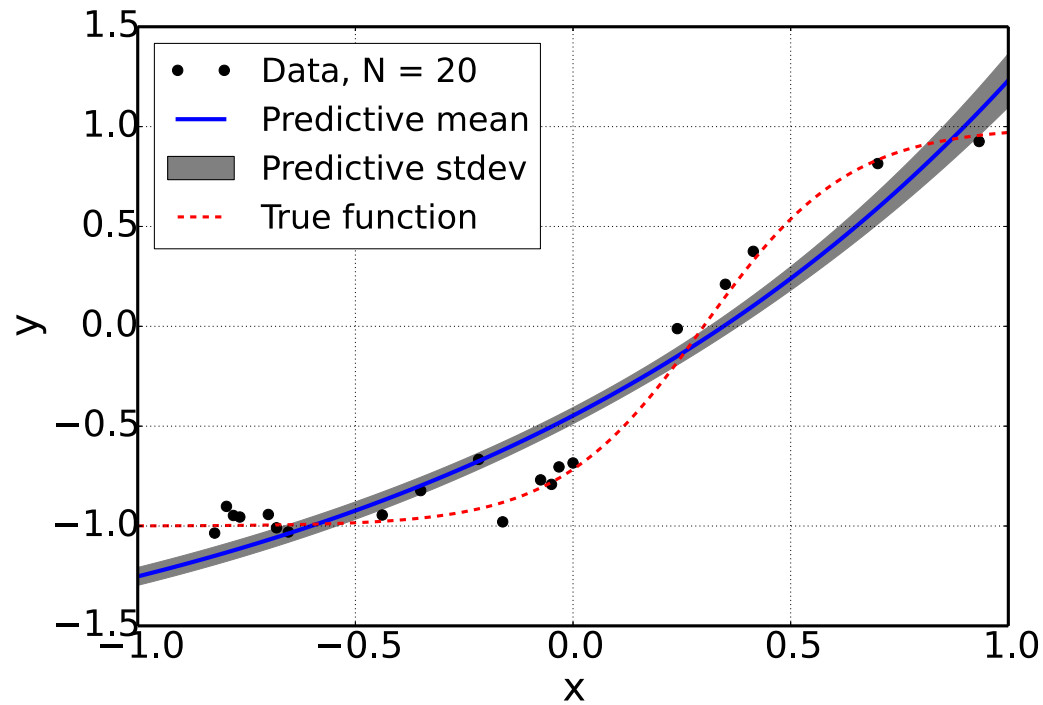


Calibrate an exponential model, but data comes from a different function (there is model error!)

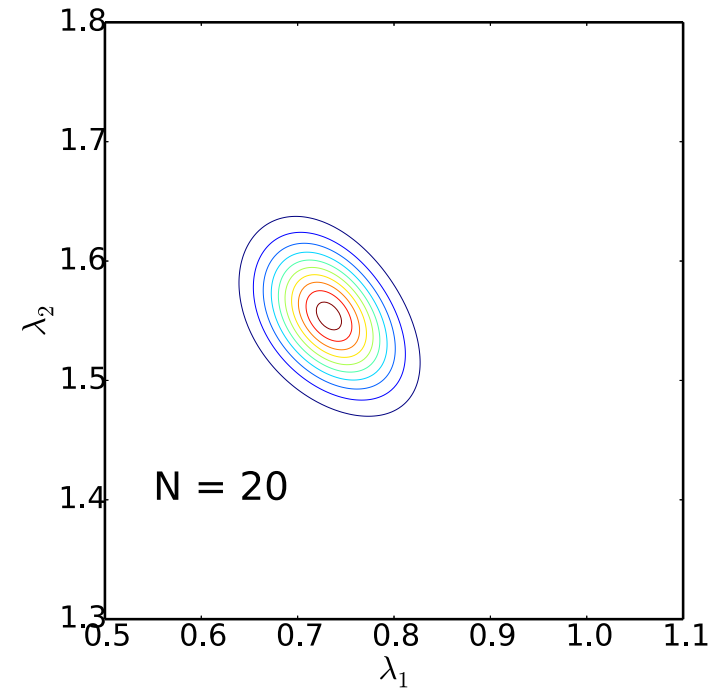
Wrong model leads to biased estimation



Model prediction vs data



Posterior PDF of model parameters

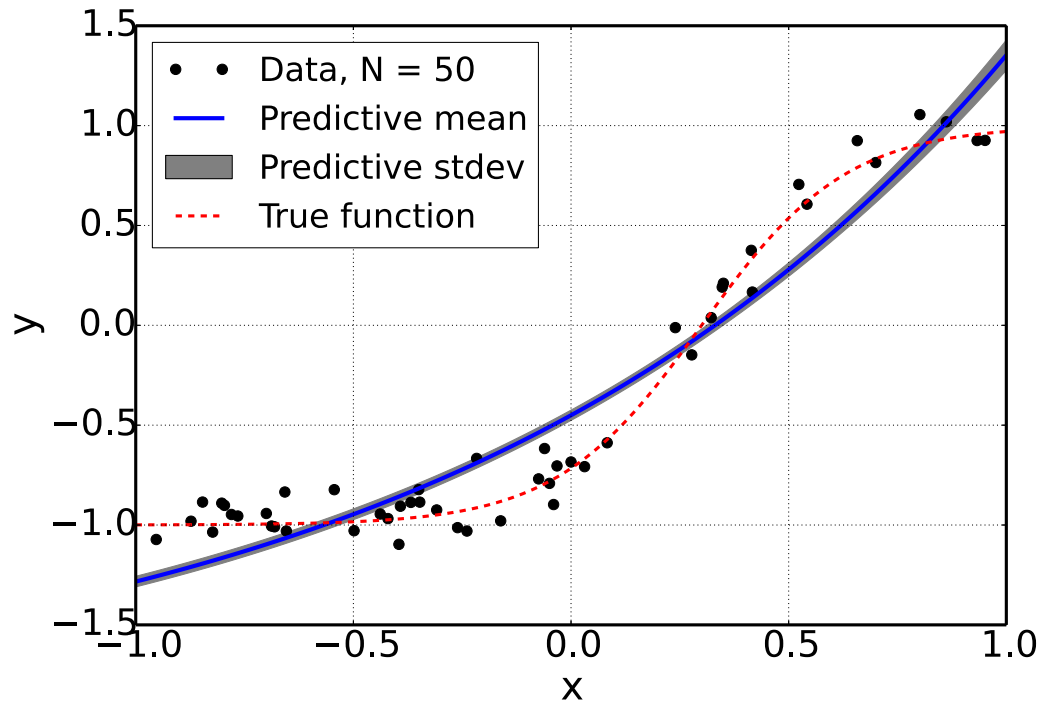


Collecting more data: become increasingly sure about the wrong values of parameters

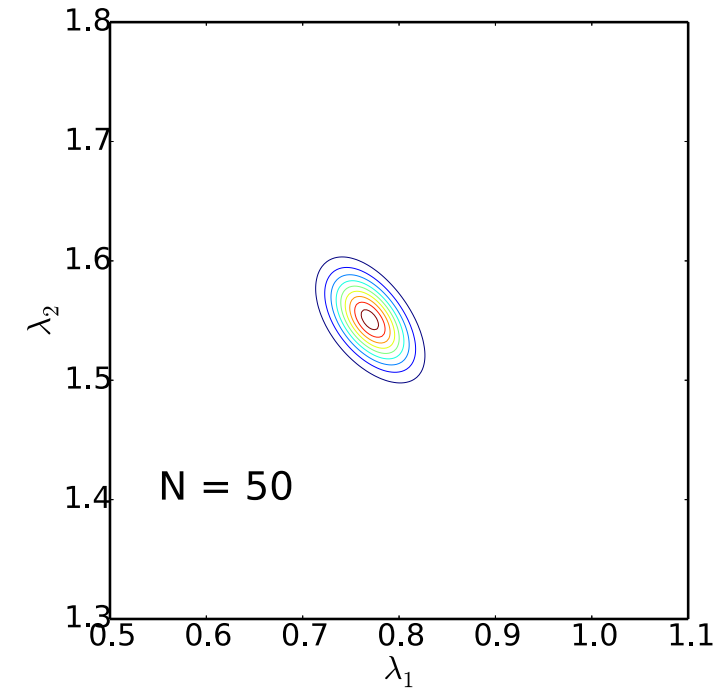
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Model prediction vs data



Posterior PDF of exponential parameters

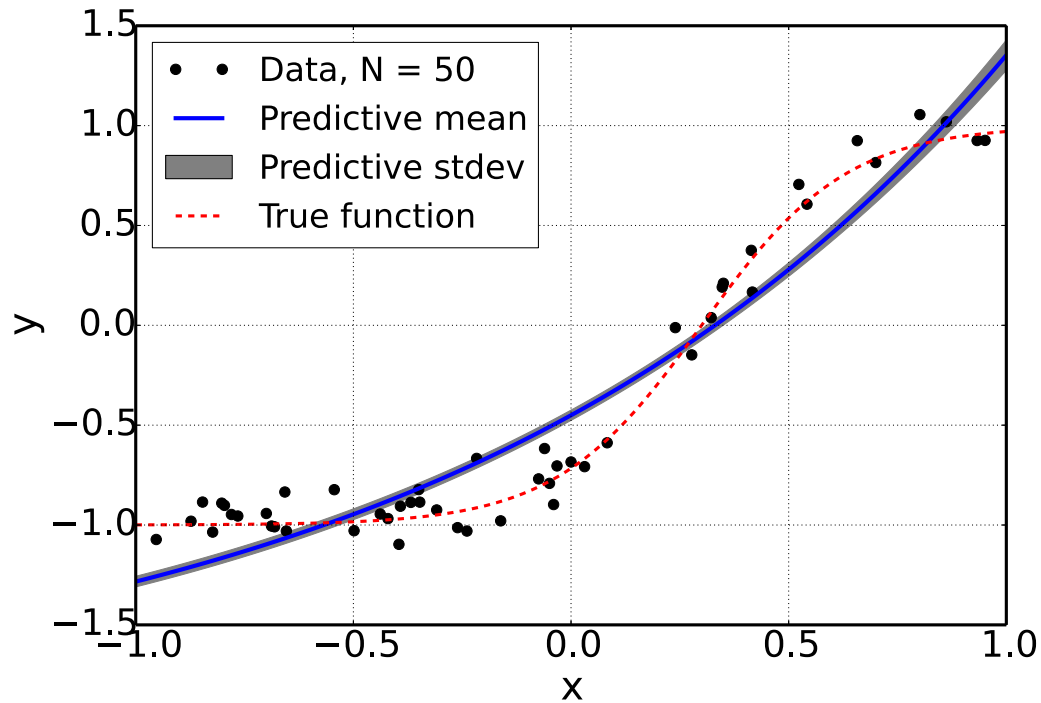


Collecting more data: become increasingly sure about the wrong values of parameters

Wrong model leads to biased estimation

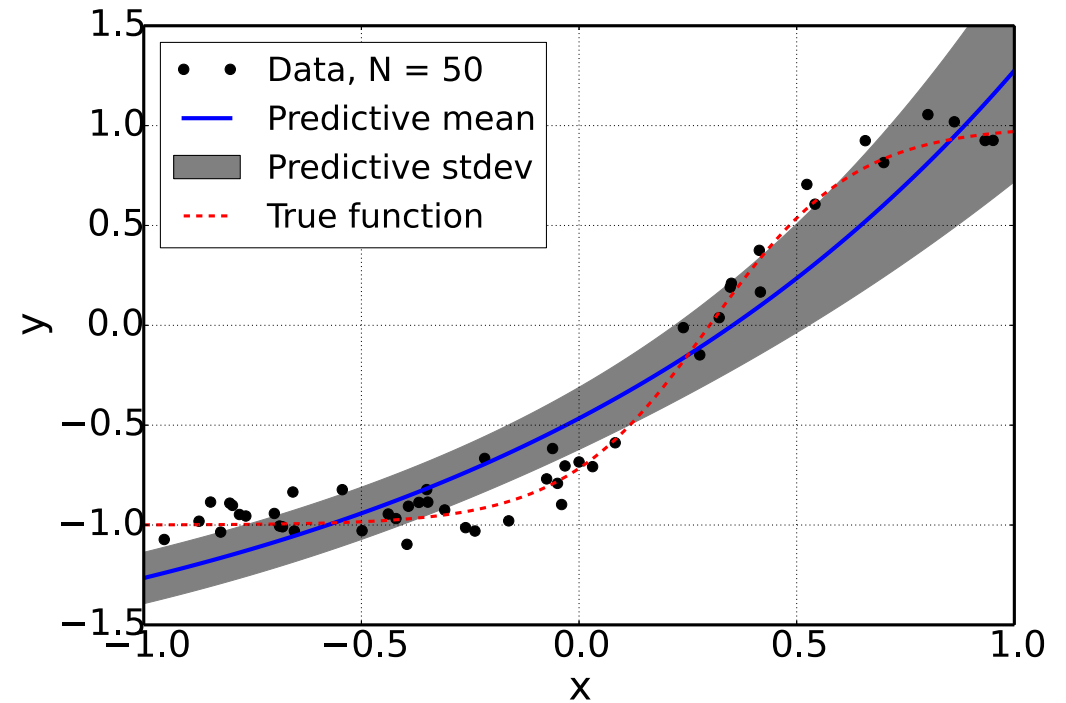


Model prediction vs data



Predictive uncertainties
do not capture
model-data discrepancy

... what we actually want



Predictive uncertainties
capture
model-data discrepancy

Embedded, non-intrusive workflow is implemented in UQTK



Funded by FASTMath SciDAC Institute, www.sandia.gov/uqtoolkit

- Model error correction inside the model, parameterized by ***polynomial chaos (PC)***
- Simultaneous ***Bayesian inference*** of physical parameters and model error parameters
- Calibrated uncertain prediction with model error component

$$y(x_i) = f(x_i; \lambda + \overset{\text{PC}}{\delta(x_i)}) + \epsilon_i$$

Bayesian inference

Applications

- Earth System Land Model (BER)
- Chemical kinetic modeling (BES)
- Plasma surface interactions (FES)
- Turbulence modeling (DARPA)

Method: Sargsyan, Najm, Ghanem, IJCK (2015); Sargsyan, Huan, Najm, IJUQ (2019).

Applications: Huan et. al, AIAA J (2018); Hakim et. al, CTM (2018); Cekmer et. al, IJUQ (2018); Rizzi et. al, CMAME (2019).

Unique capability for model exploration via **embedded** model error representation



Non-intrusive

$$y(x_i) = f(x_i; \lambda + \delta(x_i)) + \epsilon_i$$

- Allows meaningful extrapolation
 - Respects physics
 - Disambiguates model and data errors
- Predictive uncertainty attribution
- surrogate errors
 - data noise
 - parametric uncertainty
 - **structural errors**

Intrusive

$$y(x_i) = \tilde{f}(x_i; \lambda, \delta(x_i)) + \epsilon_i$$

- For best impact, always look under the hood
 - currently, we do this for the land model

- Code available via UQTK (www.sandia.gov/uqtoolkit)
- Impacted many programs DOE/DOD/SNL
- Applied outside immediate group
- Provides alternative for external correction approach currently available in DAKOTA

UQTK

Prediction variance

=

parametric uncertainty

+

data noise

+

model error